Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Abstract

[To be continued...]

- Appendix on s-wave SC ghost simulations on HM
- Set ComputerModern font in all CairoMakie figures

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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
PH	Particle-hole
PHS	Particle-hole symmetry
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

Chapter 1

The normal phase

In this chapter we apply the HF methods sketched in Sec. **??** to the "normal phase". We are studying the model in a state were no symmetry of those enumerated in Tab. **??** is broken. Even though this phase may appear not so interesting, it actually shows a significant physical difference with respect to the conventional perfectly degenerate Fermi gas.

HF methods applied in this context will lead us to a self-consistent hopping renormalisation effect. For certain values of the non-local attraction V and the doping level, a complete Mott localisation effect takes place. The resulting bands shrinking and flattening is the main net effect of the presence of $\hat{H}_V$ in the EHM.

This chapter is divided in three main parts. First, we discuss the symmetries of the normal phase, following the general discussion of Chap. **??**. Second, we apply HF methods to derive the self-consistent approximated hamiltonian and impose some theoretical constraints on the self-consistent solution. Lastly, we present numeric results.

1.1 The normal phase and its symmetries

In the normal phase no symmetry is broken. Thus, the HF wavefunction approximating the ground state must exhibit the full $\mathrm{U}(2) \otimes \mathrm{C}_4 \otimes \mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$ symmetry discussed in Sec. **??**. We take a slight detour from this, and instead allow for

$$\mathrm{C}_4 \overset{\text{SSB}}{\to} \mathrm{C}_2$$

This is the only potential SSB we consider. It will turn out in Sec. 1.5.2 that even allowing for breaking of rotational symmetry, the ground-state actually does not break C_4. This can be seen as a convolute way to confirm that there is not a normal-nematic phase. While this may seem as an useless complication, the derivation will set grounds for actual nematic effects found in the superconducting phase.

Our HF method relies on the application of Wick's theorem on the quartic part of the interacting hamiltonian, expressed in Eq. (**??**). We will decompose the quartic operators according to Eqns. (**??**), (**??**). Recalling Tabs. **??** and **??**, for the normal phase we must consider the following contributes to the EVs of Eqns. (**??**), (**??**), (**??**):

$\hat{H}_U$	Hartree contractions
$\hat{H}_V$ (o.s. sector)	Hartree contractions
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

As we will see in detail and is summarised in Tab. 1.1, Hartree contractions give a chemical potential shift, while the unique Fock contraction effectively renormalises the bare bands.

In the normal phase, spin is a pure physical degeneracy, being the $\uparrow$ and $\downarrow$ sectors separated; moreover, $\hat{H}$ is $\mathbb{Z}_2$ invariant with respect to spin inversion, $\sigma \leftrightarrow \overline{\sigma}$. This implies that all physical quantities, as is for the normal phase, should not depend on the spin index. We will perform all computations leaving σ explicit and using this consideration at the end.

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree	μ shift by nzV
		Fock	Bands renormalisation and resymmetrization

Table 1.1 | Relevant Wick's contractions and net effect in the normal phase.

1.2 Application of HF methods to the normal phase

The quartic part of $\hat{H}$ is given by the the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$. Applying HF methods to them, we get the decomposition of Eqns. (**??**), (**??**). In the next section we analyse the Hartree channel, which contains all Hartree-like terms. We will see that $\hat{H}_U$ and $\hat{H}_V$ are approximated, in this channel, by a multiple of the number operator. In the section that follows, we deal with Fock terms, and show how $\hat{H}_V$ is approximated, in that channel, by a kinetic-like operator.

1.2.1 Hartree effects: chemical potential renormalisation

Both the two-body operators $\hat{H}_U$, $\hat{H}_V$ have non-zero Hartree EVs. We deal with them separately, and show that both lead to a renormalisation of μ.

Local repulsion. In the normal phase, the local repulsion $\hat{H}_U$ decomposes in Hartree channel as specified in Eq. (**??**)

$$\hat{H}_U \overset{\text{HF}}{\simeq} U \sum_i \left(\langle \hat{n}_{i\uparrow} \rangle \hat{n}_{i\downarrow} + \hat{n}_{i\uparrow} \langle \hat{n}_{i\downarrow} \rangle - \langle \hat{n}_{i\downarrow} \rangle \langle \hat{n}_{i\uparrow} \rangle \right)$$

For a translationally invariant phase such as the normal phase, it holds

$$\langle \hat{n}_{i\sigma} \rangle = n \tag{1.1}$$

which gives

$$\begin{aligned} \hat{H}_U &\overset{\text{HF}}{\simeq} nU \sum_i (\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}) - U \sum_i n^2 \\ &= nU \times \hat{N} - E^{(\text{N})}_{\text{H}/U} \end{aligned} \tag{1.2}$$

being $\hat{N}$ the total number operator and $E^{(\text{N})}_{\text{H}/U}$ the "local Hartree contraction energy" (also known as double counting term),

$$E^{(\text{N})}_{\text{H}/U} = L_x L_y \times n^2 U \tag{1.3}$$

This energy contributes negatively to the ground-state energy, for $U > 0$. The chemical potential is corrected by

$$\mu \to \mu - nU \tag{1.4}$$

The non-local attraction further corrects μ.

Non-local attraction. The non-local attraction is divided in two sectors, as in Eq. (**??**). From Eq. (**??**), its Hartree-like terms are:

$$\hat{H}_V \overset{\text{HF}}{\simeq} \overbrace{-V \sum_{\langle ij \rangle} \sum_\sigma \left(\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma} \langle \hat{n}_{j\sigma} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle \right)}^{\text{s.s.}} \\ \underbrace{-V \sum_{\langle ij \rangle} \sum_\sigma \left(\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma} \langle \hat{n}_{j\bar{\sigma}} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle \right)}_{\text{o.s.}} + (\text{all the rest})$$

where "all the rest" collects all non-Hartree terms. Recalling the Ansatz of Eq. (1.1), we get

$$\hat{H}_V \overset{\text{HF}}{\simeq} \underbrace{-nV\sum_{\langle ij\rangle}\sum_{\sigma}(\hat{n}_{j\sigma}+\hat{n}_{i\sigma})}_{\text{s.s.}}\underbrace{-nV\sum_{\langle ij\rangle}\sum_{\sigma}(\hat{n}_{j\bar{\sigma}}+\hat{n}_{i\sigma})}_{\text{o.s.}}-E^{(\text{N})}_{\text{H}/V}+(\text{all the rest}) \tag{1.5}$$

with $E^{(\text{N})}_{\text{H}/V}$ the normal state shift to energy brought by $\hat{H}_V$:

$$\begin{aligned} E^{(\text{N})}_{\text{H}/V} &= -V\sum_{\langle ij\rangle}\sum_{\sigma}\left(\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle+\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\bar{\sigma}}\rangle\right) \\ &= -2V\sum_{\langle ij\rangle}\sum_{\sigma}n^2 \\ &= -L_xL_y\times 2zn^2V \end{aligned} \tag{1.6}$$

Note the similarity with Eq. (**??**).

Getting back to Eq. (1.5), both sums give a number operator. Note that we sum over NNs: each site is connected to $z=4$ sites. Including this combinatorial factor, we have

$$\hat{H}_V \simeq -2nzV\times\hat{N}-E^{(\text{N})}_{\text{H}/V}+(\text{all the rest}) \tag{1.7}$$

This accounts for the final Hartree shift of μ,

$$\mu \to \mu+2znV \tag{1.8}$$

Collecting everything, we get the HF approximation to the two interactions in the Hartree channel:

$$\hat{H}_U+\hat{H}_V \overset{\text{HF}}{\simeq} -n(2zV-U)\hat{N}$$

Considering the net shift to μ due to both interactions by combining Eqns. (1.4), (1.8), we get the final RMP anticipated in Tab. 1.1,

$$\tilde{\mu} \equiv \mu+n(2zV-U) \tag{1.9}$$

This result will remain valid throughout all phases discussed in this text.

1.2.2 Fock effects: bare bands renormalisation

We move to Fock terms. From Wick's decomposition of $\hat{H}_V$, the unique allowed Fock term comes from the same-spin sector. This selection rules is implied by the SU(2) symmetry of the normal phase and was stated in Tab. **??**. Then, from Eq. (**??**):

$$\hat{H}_V \overset{\text{HF}}{\simeq} (\text{Hartree channel})+V\sum_{\langle ij\rangle}\sum_{\sigma}\left(\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right) \tag{1.10}$$

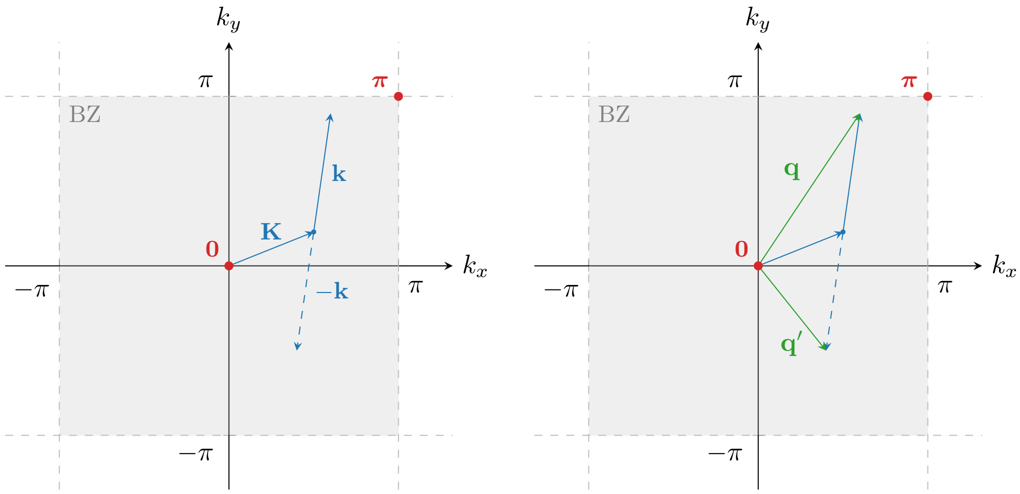
Note the plus sign in front of the displayed term: it comes from the minus sign in Fock term of Eq. (**??**).

Consider the Fourier Transform $\mathcal{F}$ given in Eq. (**??**), which gives

$$\begin{aligned} \hat{H}_V \overset{\text{HF}}{\simeq}\; & (\text{Hartree channel}) \\ & +\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left(\cos(\delta k_x)+\cos(\delta k_y)\right)\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}-E^{(\text{N})}_{\text{F}/V} \end{aligned} \tag{1.11}$$

Here, the 2 prefactor comes from the fact that the first two operators in square brackets in Eq. (1.10) are equal in reciprocal space. The double counting energy shift is given by

$$E^{(\text{N})}_{\text{F}/V} \equiv \frac{V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left(\cos(\delta k_x)+\cos(\delta k_y)\right)\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\langle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\rangle \tag{1.12}$$



a | Generic vectors.

b | Change of variables.

Figure 1.1 | Representation of the variable change employed to pass from Eq. (1.17) to (1.18). In Fig. 1.1a generic vectors are considered, cycling over all values of $\mathbf{K}, \mathbf{k} \in$ BZ. In Fig. 1.1b is depicted the variables change to the new vectors $\mathbf{q}, \mathbf{q}'$.

We define

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle \qquad u_{\mathbf{k}\sigma} = u^*_{\mathbf{k}\sigma} \tag{1.13}$$

Using the functions of Tab. **??**, we define its harmonics:

$$u^{(\gamma)}_\sigma \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle \tag{1.14}$$

The non-local attraction $\hat{H}_V$ is approximated, in Fock channel, by:

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} (\text{Hartree channel}) + V \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \left(u^{(s^*)}_\sigma \varphi^{(s^*)*}_{\mathbf{k}} + u^{(d)}_\sigma \varphi^{(d)*}_{\mathbf{k}} \right) \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{1.15}$$

We provide a proof.

Proof 1.1 (Proof of Eq. (1.15)).

The normal phase is described by a perfectly degenerate Fermi gas, hence it must hold:

$$\langle \hat{c}^\dagger_{\mathbf{k}_1\sigma_1} \hat{c}_{\mathbf{k}_2\sigma_2} \rangle = \delta_{\mathbf{k}_1=\mathbf{k}_2} \delta_{\sigma_1=\sigma_2} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \tag{1.16}$$

being f the Fermi-Dirac distribution,

$$f(\epsilon; \beta, \mu) \equiv \frac{1}{e^{\beta(\epsilon-\mu)} + 1}$$

Consider now Eq. (1.11). Applying Eq. (1.16), it becomes:

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} (\text{Hartree channel}) + \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} (\cos 2k_x + \cos 2k_y) \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{1.17}$$

Let now $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$, $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. A representation of this variables change is given in Fig. 1.1.

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = \dfrac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. **??**
$d_{x^2-y^2}$-wave	$u^{(d)} = \dfrac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. **??**

Table 1.2 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

Being (for $\ell = x, y$)

$$\begin{aligned}\delta q_\ell &\equiv q_\ell - q'_\ell \\ &= (K_\ell + k_\ell) - (K_\ell - k_\ell) \\ &= 2k_\ell\end{aligned}$$

we reduce Eq. (1.11) to

$$\begin{aligned}\hat{H}_V \overset{\mathrm{HF}}{\simeq}& \text{ (Hartree channel)} \\ &+ \frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} [\cos(\delta q_x) + \cos(\delta q_y)] \underbrace{\langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle}_{u_{\mathbf{q}\sigma}} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V}\end{aligned} \tag{1.18}$$

where we recognised $u_{\mathbf{q}\sigma}$, as was defined in Eq. (1.13). Using Eq. (**??**), we obtain

$$\begin{aligned}\hat{H}_V \overset{\mathrm{HF}}{\simeq}& \text{ (Hartree channel)} \\ &+ \sum_{\gamma,\sigma} \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} u_{\mathbf{q}\sigma} \right) \times V \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V}\end{aligned} \tag{1.19}$$

where we used the functions defined in Tab. **??**. We recognise the harmonics $u^{(\gamma)}_\sigma$ from Eq. (1.14),

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} \text{(Hartree channel)} + V \sum_{\mathbf{q}'\in\mathrm{BZ}} \sum_{\gamma,\sigma} u^{(\gamma)}_\sigma \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{1.20}$$

which is almost Eq. (1.15).

To complete the proof, we recognise that in the normal phase inversion symmetry is not broken, thus

$$u_{\mathbf{q}\sigma} = u_{-\mathbf{q}\sigma}$$

This follows from the fact that in the normal phase the EV of the operator $\hat{n}_{\mathbf{k}\sigma}$ must be inversion-symmetric. It follows that antisymmetric harmonics are zero,

$$u^{(p_\ell)}_\sigma = 0 \quad \text{for} \quad \ell \in \{x, y\}$$

Using this observation in Eq. (1.20), we obtain Eq. (1.15).

Composing Eq. (1.15) with Eqns. (1.2), (1.7) we obtain the quadratic HF approximation to the EHM, for the normal phase. The chemical potential $\tilde{\mu}$ is fixed by density and determined during computations. To find the ground-state, we must determine self-consistently the two HFPs $u^{(s^*)}_\sigma$, $u^{(d)}_\sigma$ through the self-consistency Eq. (1.14). The symmetry-resolved equations are reported in Tab. 1.2.

The operator of Eq. (1.15) has the same algebraic structure of the kinetic operator $\hat{H}_t$ in reciprocal space, given in Eq. (**??**). Hence, we can collapse the two into one effective kinetic operator. Let

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u^{(s^*)}_\sigma V(\cos k_x + \cos k_y) + u^{(d)}_\sigma V(\cos k_x - \cos k_y) \tag{1.21}$$

be the renormalised bare bands. In Fock channel, we have:

$$\hat{H}_t + \hat{H}_V \overset{\text{HF}}{\simeq} (\text{Hartree channel}) + \underbrace{\sum_{\mathbf{k}\in\text{BZ}} \sum_{\sigma} \tilde{\epsilon}_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}}_{\hat{H}_{\tilde{t}}} - E^{(\text{N})}_{\text{F}/V}$$

where we defined the effective kinetic operator, $\hat{H}_{\tilde{t}}$. The renormalised bands of Eq. (1.21) are a sum of s^* and d-wave components. Instead, bare bands as defined Eq. (**??**) are purely s^*-symmetric functions, $\epsilon_{\mathbf{k}\sigma} = -4t\varphi^{(\gamma)}_{\mathbf{k}}$.

In real space, $\hat{H}_{\tilde{t}}$ is given by:

$$\begin{aligned} \hat{H}_{\tilde{t}} &= \hat{H}_t + V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \text{h.c.} \right] \\ &= - \sum_{\langle ij \rangle} \sum_{\sigma} \left[\tilde{t}_{ij\sigma} \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] \end{aligned}$$

where we defined the effective hopping amplitude

$$\tilde{t}_{ij\sigma} \equiv t - V \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \qquad (\text{for } i, j \text{ NNs})$$

The renormalised hopping $\tilde{t}_{ij\sigma}$ is related via a Fourier transform to the renormalised bands $\tilde{\epsilon}_{\mathbf{k}\sigma}$.

We conclude this section by proving that the Fock contraction energy is given by:

$$E^{(\text{N})}_{\text{F}/V} = L_x L_y \times \frac{V}{2} \sum_{\sigma} \left(\left| u^{(s^*)}_{\sigma} \right|^2 + \left| u^{(d)}_{\sigma} \right|^2 \right) \tag{1.22}$$

Proof 1.2 (Non-local Fock contraction energy.).

Recall Eq. (1.12). Using Eq. (1.16), it reduces to:

$$E^{(\text{N})}_{\text{F}/V} = \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos(\delta q_x) + \cos(\delta q_y) \right] \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \langle \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} \rangle$$

Inserting Eq. (1.13), we get:

$$\begin{aligned} E^{(\text{N})}_{\text{F}/V} &= L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} u_{\mathbf{q}\sigma} \right) \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in\text{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} u^*_{\mathbf{q}'\sigma} \right) \\ &= L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left| u^{(\gamma)}_{\sigma} \right|^2 \end{aligned}$$

which, using the fact that $u^{(p_\ell)}_{\sigma} = 0$, proves Eq. (1.22).

In next section we collect all we found so far, discuss the properties of the HF hamiltonian and derive the free energy and entropy densities for its ground-state.

FROM HERE...

1.3 The HF hamiltonian and its properties

This equation defines the bands renormalisation anticipated in Tab. 1.1, and is actually spin-independent. The fact that we have both s^* and a d-wave components introduces an anisotropy that is discussed in next section. However, we anticipate here that, as will result from Sec. 1.5.2, the d-wave harmonic will be suppressed. For now we keep this component ignoring these anticipations.

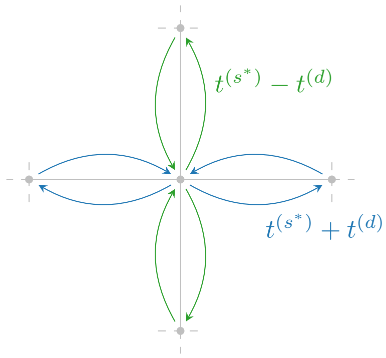


Figure 1.2 Graphic rendition of anisotropic hopping, with a s^*-wave component (symmetric by $\pi/$ rotations) and a d-wave component (antisymmetric).

1.3.1 Nematicity: extending hopping to d-wave bands

For the normal phase, there is no reason to expect nematic effects, nor to exclude them. They are observed in superconducting phase [23]. Thus, we allow the possibility.

From Tabs. **??** and **??** we can inverse-Fourier transform the form factors to get the full effective hopping,

$$\tilde{t}_{ij\sigma} = \sum_{\gamma \in \{s^*, d\}} t_\sigma^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

with

$$t_\sigma^{(s^*)} = t - \frac{u_\sigma^{(s^*)} V}{2} \qquad t_\sigma^{(d)} = -\frac{u_\sigma^{(d)} V}{2}$$

This is what we intend for "nematic effects". The possibility that planar rotational symmetry is broken in such a way that the kinetic operators in x and y directions differ. A schematics of anisotropic hopping is given in Fig. 1.2: for each bond linking two sites, the hopping parameter can be a positive or negative combination of the two components, whether the link direction is vertical or horizontal. The solution needs to be invariant under exchange $x \leftrightarrow y$, thus the actual SSB is $\mathrm{C}_4 \overset{\text{SSB}}{\to} \mathrm{C}_2/\mathbb{Z}_2$.

The fact that hopping is here anisotropic does not break full translational invariance $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$. Imposing a free degenerate Fermi gas ground state in Eq. (1.16) leads to a uniform density. This is proven by Fourier-transforming the site number operator according to the convention of Eq. (**??**),

$$\hat{c}^\dagger_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{k}, \mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma}$$

Then the on-site density EV is

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} f(\tilde{\epsilon}_\mathbf{k}; \beta, \tilde{\mu})$$

A perfectly degenerate Fermi gas is always translational invariant and thus cannot exhibit non-uniform symmetry. The fact that hopping is allowed to be anisotropic does not interfere with that.

Note that for null HFPs,

$$u_\sigma^{(s^*)} = 0 \qquad u_\sigma^{(d)} = 0$$

we have, recalling Tab. **??**:

$$\tilde{t}_{\mathbf{r}\mathbf{r}'\sigma} = \frac{t}{2} \left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}} \right]$$

The fact that $t/2$ appears here instead of t is a consequence of the equivalence:

$$-t \sum_{\langle ij \rangle} \sum_\sigma \left[\hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] = -\frac{t}{2} \sum_{i,j \in \mathcal{S}} \sum_\sigma \left[\hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right]$$

where we including a prefactor that corrects double-counting. The RMP $\tilde{t}_{ij\sigma}$ must be interpreted like the right-hand side of the above equation, with the kinetic operator becoming:

$$\hat{H}_{\tilde{t}} \equiv -\frac{1}{2}\sum_{i,j\in\mathcal{S}}\sum_{\sigma}\left[\tilde{t}_{ij\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma} + \text{h.c.}\right]$$

To limit the sum to NNs is a matter of the symmetry structure of the function $\tilde{t}_{ij\sigma}$.

Recall the functions of Tab. **??**. We have the identities:

$$\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} = \varphi^{(s^*)}_{\mathbf{rr}'} + \varphi^{(d)}_{\mathbf{rr}'}$$
$$\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}} = \varphi^{(s^*)}_{\mathbf{rr}'} - \varphi^{(d)}_{\mathbf{rr}'}$$

Then we can define the renormalised hopping on each direction as:

$$\tilde{t}^{(x)} \equiv t - \frac{u^{(s^*)} + u^{(d)}}{2}V$$
$$\tilde{t}^{(y)} \equiv t - \frac{u^{(s^*)} - u^{(d)}}{2}V$$

Both components depend linearly on V. The HFP $u^{(s^*)}$ is a measure of the direction-averaged hopping shift with respect to V, while $u^{(d)}$ measures the anisotropy,

$$u^{(s^*)} = \frac{\left(t - \tilde{t}^{(x)}\right) + \left(t - \tilde{t}^{(y)}\right)}{V} \tag{1.23}$$
$$u^{(d)} = \frac{\tilde{t}^{(x)} - \tilde{t}^{(y)}}{V} \tag{1.24}$$

It follows that if $u^{(d)} > 0$, the hopping in direction x is boosted while in direction y is damped. If $u^{(d)} < 0$, the contrary.

This discussion is general, and not specific to the normal phase. In next chapters we will interpret phase renormalisation as in this section.

1.4 Normal free energy density

In the end, we want to compare phases. Thus we need a way to extract the free energy density of the approximated ground-state for each phase. The procedure sketched here will be re-used in next chapters. Let

$$f^{(\text{N})} \equiv \frac{F^{(\text{N})}_{\text{MFT}}}{L_x L_y}$$

being F_{MFT} the total free energy. Consider the complete hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k},\sigma}\tilde{\epsilon}_{\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} + E^{(\text{N})}_{\text{c}} - \tilde{\mu}\hat{N}$$

where $E^{(\text{N})}_{\text{c}}$ is the (negative) total contraction energy for the normal phase,

$$E^{(\text{N})}_{\text{c}} \equiv -\left[E^{(\text{N})}_{\text{H}/U} + E^{(\text{N})}_{\text{H}/V} + E^{(\text{N})}_{\text{F}/V}\right]$$

having included the various contraction energies from Eqns. (1.3), (1.6) and (**??**).

Let us now take together all terms. From now on, we will omit spin index σ, making use of the always present $\mathbb{Z}_2$ spin inversion symmetry discussed in Sec. 1.1. For a free Fermi gas, the free energy density is made of four contributions:

1. The free particles ground state energy density,

$$e^{(\text{N})}_{\text{g}} = \frac{2}{L_x L_y}\sum_{\mathbf{k}\in\text{BZ}}(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

2. The contraction energy density,

$$e_{\mathrm{c}}^{(\mathrm{N})} \equiv \frac{E_{\mathrm{c}}^{(\mathrm{N})}}{L_x L_y} = -n^2 U + V\left[2zn^2 - \left|u^{(s^*)}\right|^2 - \left|u^{(d)}\right|^2\right]$$

3. The entropic contribution to free energy

$$e_s^{(\mathrm{N})} \equiv -Ts$$

with $T = 1/k_B\beta$ and s the free Fermi gas entropy density for a single spin sector [29],

$$s = -\frac{k_B}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}))\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right] \tag{1.25}$$

We use the identity:

$$\ln\left(1 - \frac{1}{e^x + 1}\right) = x + \ln\left(\frac{1}{e^x + 1}\right)$$

and, substituting $x = \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})$, reduce the above expression to:

$$s = -\frac{k_B}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\Big[\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) - \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\Big]$$

Second and fourth terms cancel out, and multiplying by $-2T$ (taking care of spin degeneracy) we have

$$e_s^{(\mathrm{N})} = \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - \frac{2}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

4. The chemical potential correction

$$e_n^{(\mathrm{N})} = \mu n$$

Note here that we are using the "physical" value of μ, not its RMP counterpart $\tilde{\mu}$. The reason for this is discussed in Sec. **??**.

Thus, composing everything

$$f^{(\mathrm{N})} \equiv e_{\mathrm{g}}^{(\mathrm{N})} + e_{\mathrm{c}}^{(\mathrm{N})} + e_s^{(\mathrm{N})} + e_n^{(\mathrm{N})} = \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + \left[\mu + n(2zV - U)\right]n$$

In Sec. **??** we argued that our iterative scheme computes $\tilde{\mu}$, not μ. Using Eq. (1.9), we see

$$\mu + n(2zV - U) = \tilde{\mu}$$

Then, the final expression is:

$$f^{(\mathrm{N})} = \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + \tilde{\mu}n \tag{1.26}$$

including all effects.

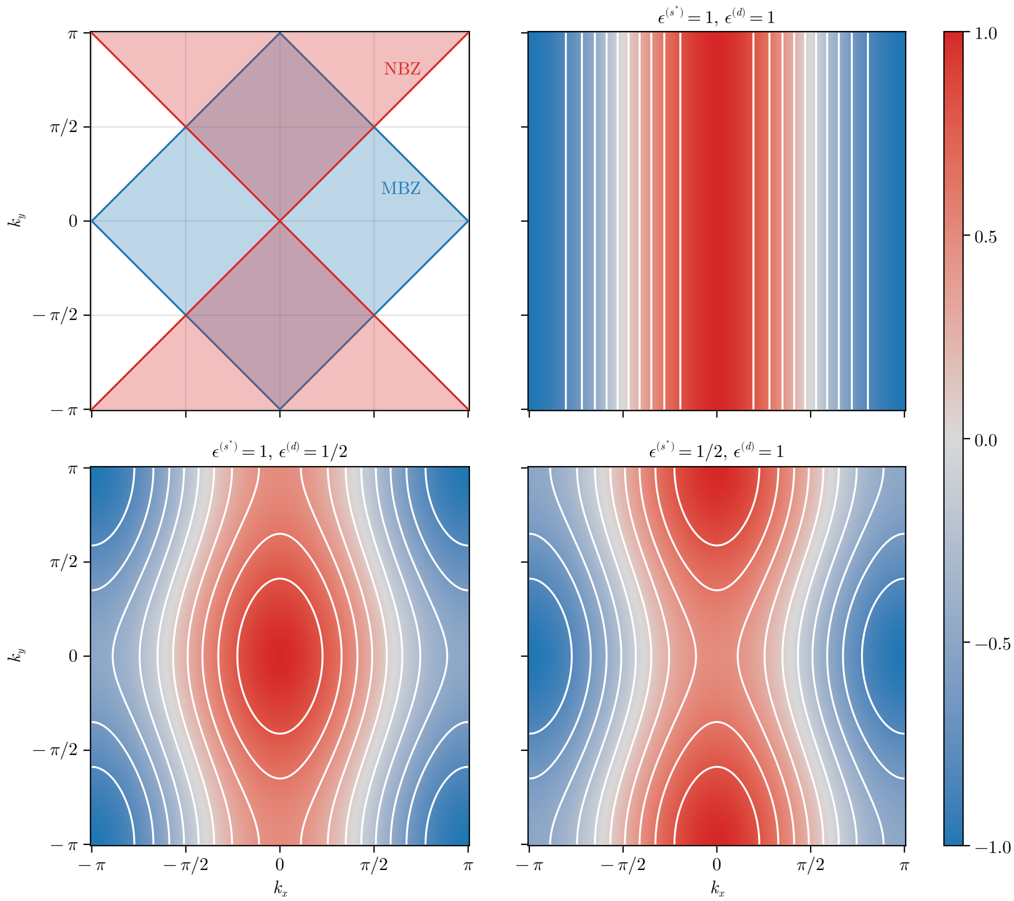


Figure 1.3 Theoretical representation of mixed-symmetry nematic bands for the normal phase. The white contour represent level lines. In top-left panel, the MBZ and NBZ are depicted. While these plots only account for positive-defined components, in text is described how to map each one on its counterpart with different sign.

1.5 Theoretical constraints on the normal phase HFPs

[Move this section to appendix?]

Within this section we discuss some theoretical constraints on the two HFPs of the normal phase. The discussion is divided in two parts: first we consider the simpler case of unviolated C_4 symmetry, then we move to anisotropic hopping and $C_4 \overset{\text{SSB}}{\to} C_2$.

Before starting, it is useful to spend a few words on the spatial structure of a mixed $s^* \otimes d$-wave band. Let in general for notational simplicity

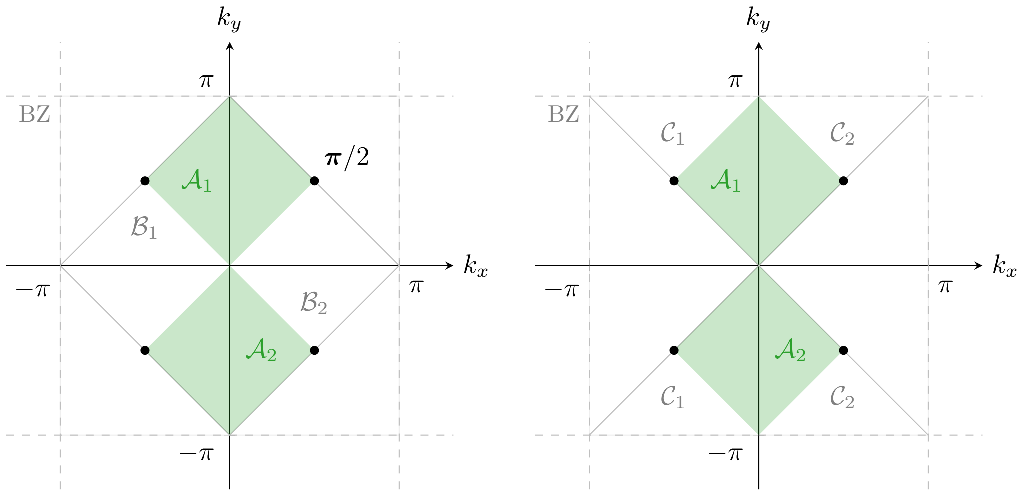
$$\tilde{\epsilon}_{\mathbf{k}} = \epsilon^{(s^*)}(\cos k_x + \cos k_y) + \epsilon^{(d)}(\cos k_x - \cos k_y)$$

It is the balance between the two components $\epsilon^{(s^*)}$ and $\epsilon^{(d)}$ that gives the main features of the renormalised band. Consider the s^*-wave and d-wave structure factors already presented in Fig. **??**, and define the Magnetic Brillouin Zone (MBZ) and Nematic Brillouin Zone (NBZ) as follows:

$$\text{MBZ} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \right\} \tag{1.27}$$

$$\text{NBZ} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \tag{1.28}$$

represented in the top-left panel of Fig. 1.3. The reason for the name of the MBZ will be clear in Chap. **??**. In the rest of the panels of Fig. 1.3 are reported examples of mixed symmetry bands,



a | MBZ sectors. b | NBZ sectors.

Figure 1.4 | Depiction of the sectors composing the MBZ (Fig. 1.4a) and the NBZ (Fig. 1.4b), based on the divisions shown in the top-left panel of Fig. 1.3. The full green zone composes $\mathcal{A}$, the region where both the s^*-wave and d-wave structure factors are positive defined; its MBZ complementary white counterpart in the left panel gives $\mathcal{B}$, for which the s^*-wave structure factors is positive-definite while the d-wave is negative. Similarly the NBZ complementary white counterpart in the right panel gives $\mathcal{C}$, for which signs are exchanged. The black dots, at the contours intersection, are band nodes.

for positive components. The respective opposite sign plots are obtained by the following rules:

$$\begin{aligned}
&\epsilon^{(s^*)} < 0 \quad \text{and} \quad \epsilon^{(d)} < 0 \quad \Longrightarrow \quad \text{Invert colors in Fig. 1.3} \\
&\epsilon^{(s^*)} > 0 \quad \text{and} \quad \epsilon^{(d)} < 0 \quad \Longrightarrow \quad \text{Rotate plots by } \pi/2 \text{ in Fig. 1.3} \\
&\epsilon^{(s^*)} < 0 \quad \text{and} \quad \epsilon^{(d)} > 0 \quad \Longrightarrow \quad \text{Rotate plots by } \pi/2 \text{ and invert colors in Fig. 1.3}
\end{aligned}$$

Let us focus on the MBZ and the NBZ. Starting from the intersections of the two zones, both can be factored in two sectors of interest,

$$\mathrm{MBZ} = \mathcal{A} \cup \mathcal{B} \qquad \mathrm{NBZ} = \mathcal{A} \cup \mathcal{C}$$

being, accordingly to Fig. 1.4,

$$\mathcal{A} \equiv \mathcal{A}_1 \cup \mathcal{A}_2 \qquad \mathcal{B} \equiv \mathcal{B}_1 \cup \mathcal{B}_2 \qquad \mathcal{C} \equiv \mathcal{C}_1 \cup \mathcal{C}_2$$

These domains satisfy:

$$\begin{aligned}
\mathcal{A} &\equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \qquad (1.29)\\
&= \mathrm{MBZ} \cap \mathrm{NBZ}
\end{aligned}$$

$$\begin{aligned}
\mathcal{B} &\equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \leq 0 \right\} \qquad (1.30)\\
&= \mathrm{MBZ} \setminus \mathrm{NBZ}
\end{aligned}$$

$$\begin{aligned}
\mathcal{C} &\equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \leq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \qquad (1.31)\\
&= \mathrm{NBZ} \setminus \mathrm{MBZ}
\end{aligned}$$

and will be used later on. Whatever the shape of the mixed band, the following equation always holds

$$|k_x| = \frac{\pi}{2} \quad \text{and} \quad |k_y| = \frac{\pi}{2} \quad \Longrightarrow \quad \tilde{\epsilon}_{(k_x,k_y)} = 0$$

since only cosines are involved. This identifies four node points, represented as black dots in Fig. 1.4.

1.5.1 Constraints in presence of a robust C_4 symmetry and bands shrinking

Consider the simpler situation where C_4 symmetry is imposed. This implies that the renormalised bands are

$$\tilde{\epsilon}_{\mathbf{k}} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y)$$

with a single self-consistency equation

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

The following chain of deductions leads us to a constraint on the possible values of $u^{(s^*)}$:

1. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \leq 0$. Let us prove this: suppose instead $u^{(s^*)} \leq 0$. For any function $g_{\mathbf{k}}$ defined on the BZ:

$$\sum_{\mathbf{k}\in\mathrm{BZ}} g_{\mathbf{k}} = \sum_{\mathbf{k}\in\mathrm{MBZ}} [g_{\mathbf{k}} + g_{\mathbf{k}+\boldsymbol{\pi}}]$$

where we defined the vector $\boldsymbol{\pi} \equiv (\pi, \pi)$. The self-consistency equation thus reads

$$\begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathbf{k}+\boldsymbol{\pi}}; \beta, \tilde{\mu}) \right] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned} \tag{1.32}$$

having we used the antisymmetry of s^*-wave functions by $\boldsymbol{\pi}$ shifts. Now, if $u^{(s^*)} \leq 0$ the renormalised band $\tilde{\epsilon}_{\mathbf{k}}$ is negative inside the MBZ and positive outside, implying

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \geq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

the equality only being satisfied on the MBZ contour. Then, since inside the MBZ it holds by definition $\cos k_x + \cos k_y \geq 0$, the above equations imply

$$u^{(s^*)} \leq 0 \quad \Longrightarrow \quad u^{(s^*)} \geq 0$$

To exclude the possibility of $u^{(s^*)} = 0$, it suffices to notice that from the argument above such a situation only arises when the MBZ contour degenerates to the entirety of the MBZ itself – which is, when the band is flat. But that cannot be the case for $u^{(s^*)} = 0$, as long as $t > 0$. Thus we have proven that

$$u^{(s^*)} > 0$$

2. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \geq 2t/V$. We proceed exactly as did for the previous point, up to the passage:

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right]$$

Now, if $u^{(s^*)} \geq 2t/V$ this means the prefactor in front of the renormalised bands is positive. Then $\tilde{\epsilon}_{\mathbf{k}}$ is positive inside the MBZ and negative outside, implying

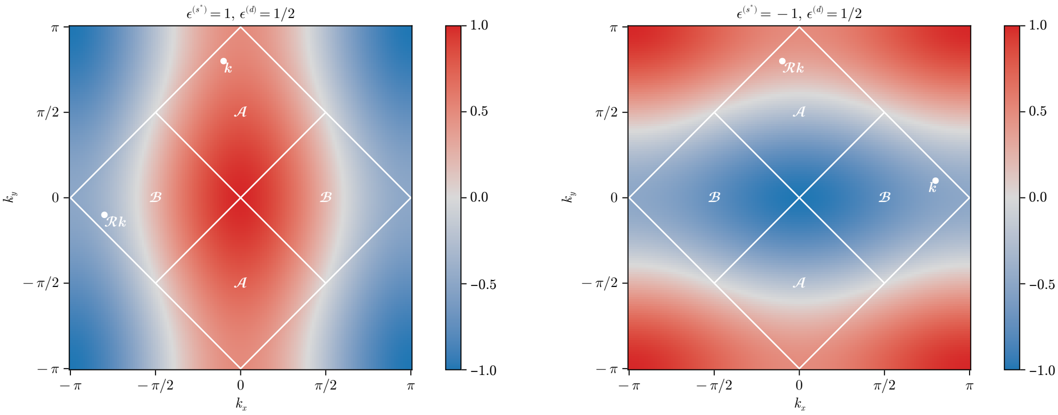
$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \leq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

thus we have

$$u^{(s^*)} \geq 2t/V \quad \Longrightarrow \quad u^{(s^*)} \leq 0$$

which is absurd. Thus

$$u^{(s^*)} < 2t/V$$



a | Same-sign mixed bands.

b | Opposite-sign mixed bands.

Figure 1.5 | Theoretical example of bands with $u^{(s^*)} > 2t/V$, $u^{(d)} > 0$ (top-left) and $u^{(s^*)} < 0$, $u^{(d)} > 0$ (top right). Two example points are reported, connected by a $\pi/2$ rotation as given in Eqns. (1.36), (1.34).

Collecting the two points above, we have the constraint on $u^{(s^*)}$

$$\mathrm{C}_4 \quad \Longrightarrow \quad 0 < u^{(s^*)} < 2t/V \tag{1.33}$$

The effective hopping is smaller than the original hopping. We can denominate this net effect "bands shrinking".

1.5.2 Natural suppression of nematicity within $\mathrm{C}_2/\mathbb{Z}_2$ symmetry

Let us extend the arguments above to a weaker symmetry, allowing for

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2/\mathbb{Z}_2$$

We work with the full, anisotropic band of Eq. (1.21). From the following chain of deductions we conclude that C_4 is not broken:

1. Let for instance

$$u^{(s^*)} \geq 2t/V > 0 \qquad u^{(d)} > 0 \quad \Longrightarrow \quad \epsilon^{(s^*)} \geq 0 \qquad \epsilon^{(d)} > 0$$

In Fig. 1.5a an example band is shown for $\epsilon^{(s^*)} = 1$, $\epsilon^{(d)} = 1/2$. Note that if $u^{(s^*)} = 2t/V$, the renormalised band is purely d-wave since $\epsilon^{(s^*)} = 0$.

Recall Eq. (1.32), and separate in two sums over $\mathcal{A}$ and $\mathcal{B}$:

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathcal{A}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\right] + \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathcal{B}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\right]$$

Consider the sum over $\mathcal{B}$. Looking at the bottom-left panel of Fig. 1.3, such a region contains nodal curves across which bands change sign. Let $\mathcal{R}$ be the clockwise $\pi/2$ rotation: then, any point in $\mathcal{B}$ can be seen as the rotation of a point in $\mathcal{A}$

$$\forall \mathbf{k}' \in \mathcal{B} \quad \exists \quad \mathbf{k} \in \mathcal{A} \mid \mathbf{k}' = \mathcal{R}\mathbf{k} \tag{1.34}$$

A graphical rendition of this idea is given in Fig. 1.5a. Then we can rewrite $u^{(s^*)}$ as a single sum over $\mathcal{A}$,

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathcal{A}} (\cos k_x + \cos k_y) \times \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu})\right] \tag{1.35}$$

From Eq. (1.29), $\forall \mathbf{k} \in \mathcal{A}$ the form factors $\cos k_x \pm \cos k_y$ are positive by construction. By hypothesis $\epsilon^{(s^*)} \geq 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)}\left(\cos k_x + \cos k_y\right)}^{\geq 0} + \overbrace{\epsilon^{(d)}\left(\cos k_x - \cos k_y\right)}^{\geq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)}\left(\cos k_x + \cos k_y\right)}_{\geq 0} - \underbrace{\epsilon^{(d)}\left(\cos k_x - \cos k_y\right)}_{\geq 0} \end{aligned}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \geq 0$$

thus $\tilde{\epsilon}_{\mathbf{k}} \geq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\leq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\geq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

This condition, inserted in Eq. (1.35), leads to an absurd:

$$u^{(s^*)} \geq 2t/V \qquad u^{(d)} > 0 \qquad \Longrightarrow \qquad u^{(s^*)} \leq 0$$

2. An identical argument, exchanging the roles of $\mathcal{A}$ and $\mathcal{B}$, can be carried out to show that

$$u^{(s^*)} \geq 2t/V > 0 \qquad u^{(d)} < 0 \qquad \Longrightarrow \qquad u^{(s^*)} \leq 0$$

3. The two points above imply that it is possible to have $u^{(s^*)} \geq 2t/V$ if and only if $u^{(d)} = 0$. But from Sec. 1.5.1, we conclude that the upper boundary $u^{(s^*)} < 2t/V$ always holds.

4. Let for instance

$$u^{(s^*)} \leq 0 \qquad u^{(d)} > 0 \qquad \Longrightarrow \qquad \epsilon^{(s^*)} < 0 \qquad \epsilon^{(d)} > 0$$

In Fig. 1.5b an example band is shown for $\epsilon^{(s^*)} = -1$, $\epsilon^{(d)} = 1/2$.

Proceed as for the first point. Similarly to Eq. (1.34), Any point in $\mathcal{A}$ can be obtained by rotating some point in $\mathcal{B}$,

$$\forall \mathbf{k}' \in \mathcal{A} \quad \exists \quad \mathbf{k} \in \mathcal{B} \mid \mathbf{k}' = \mathcal{R}\mathbf{k} \tag{1.36}$$

An equation similar to Eq. (1.35) holds, the sum being performed over $\mathcal{B}$

$$\begin{aligned} u^{(s^*)} = &\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{B}} (\cos k_x + \cos k_y) \\ &\times \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu})\right] \end{aligned} \tag{1.37}$$

From Eq. (1.30), $\forall \mathbf{k} \in \mathcal{B}$ the form factor $\cos k_x + \cos k_y$ is positive by construction; the form factor $\cos k_x - \cos k_y$ is negative by construction. By hypothesis $\epsilon^{(s^*)} < 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)}\left(\cos k_x + \cos k_y\right)}^{\leq 0} + \overbrace{\epsilon^{(d)}\left(\cos k_x - \cos k_y\right)}^{\leq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)}\left(\cos k_x + \cos k_y\right)}_{\leq 0} - \underbrace{\epsilon^{(d)}\left(\cos k_x - \cos k_y\right)}_{\leq 0} \end{aligned}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \leq 0$$

thus $\tilde{\epsilon}_{\mathbf{k}} \leq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{aligned} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\geq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\leq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

This condition, inserted in Eq. (1.35), leads to an absurd:

$$u^{(s^*)} \leq 0 \qquad u^{(d)} > 0 \qquad \Longrightarrow \qquad u^{(s^*)} > 0$$

5. can be carried out to show that

$$u^{(s^*)} \leq 0 \qquad u^{(d)} < 0 \quad \Longrightarrow \quad u^{(s^*)} > 0$$

6. The two points above imply that it is possible to have $u^{(s^*)} \leq 0$ if and only if $u^{(d)} = 0$. But from Sec. 1.5.1, we conclude that the lower boundary $u^{(s^*)} > 0$ always holds.

From the chain of derivations above, we can conclude self-consistently that also in the nematic structure the constraint of Eq. (1.33),

$$0 < u^{(s^*)} < 2t/V$$

holds. In the s^*-wave channel, partial breaking of rotational symmetry is not enough to modify the main effect identified under C_4 symmetry, "bands shrinking". What is missing is the behaviour of the d-wave channel when said effect takes place. Continuing the enumeration above,

7. Let $0 < u^{(s^*)} < 2t/V$ and $u^{(d)} > 0$. Then $\epsilon^{(d)} > 0$. The self-consistency equation for $u^{(d)}$ reads

$$\begin{aligned} u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{NBZ}} (\cos k_x - \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned}$$

   where we recognised, as is shown in Fig. 1.4b, that BZ $\setminus$ NBZ $= \mathcal{R}$NBZ. It holds:

$$\forall \mathbf{k} \in \mathrm{NBZ} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\geq 0} \end{aligned}$$

   thus $\tilde{\epsilon}_{\mathcal{R}\mathbf{k}} < \tilde{\epsilon}_{\mathbf{k}}$, and inserting this fact in the self-consistency equation above we get

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} > 0 \quad \Longrightarrow \quad u^{(d)} < 0$$

   which is absurd.

8. An identical argument, with inverted inequalities, can be carried out to show that

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} < 0 \quad \Longrightarrow \quad u^{(d)} > 0$$

9. The two points above imply that it is possible to satisfy the constraint $0 < u^{(s^*)} < 2t/V$ if and only if $u^{(d)} = 0$.

All the deductions above indicate that $u^{(d)} = 0$ everywhere. Nematic effects are always absent,

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0 \tag{1.38}$$

This is physically expected but non trivial. The normal phase does not break C_4 and we can ignore $u^{(d)}$ at all. In next section we identify the expected self-consistent value for $u^{(s^*)}$.

1.5.3 Expected dependence on δ at low temperature

We understood that $u^{(d)} = 0$. Moreover, the unique HFP $u^{(s^*)}$ is independent of U, but in general it will depend in some way on the doping level. We work at low-temperature, $\beta \to \infty$. Consider the self-consistency equation of Tab. 1.2 as a function of temperature and chemical potential,

$$u^{(s^*)}(\beta, \tilde{\mu}) = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

in the zero-temperature limit, as long as the band is not flat, it holds

$$\lim_{\beta\to\infty} u^{(s^*)}(\beta,\tilde\mu) = \frac{1}{L_xL_y}\sum_{\epsilon_{\mathbf{k}}\le\tilde\mu}(\cos k_x+\cos k_y)$$

Let $\mathrm{FS}(\delta)$ be the Fermi surface for a given doping level δ. We denote with $\mathrm{Int}(\mathrm{FS})$ be its internal part, the Fermi sea:

$$\mathrm{FS} = \{\mathbf{k}\in \mathrm{BZ} \mid \tilde\epsilon_{\mathbf{k}} = \tilde\mu\}$$
$$\mathrm{Int}(\mathrm{FS}) = \{\mathbf{k}\in \mathrm{BZ} \mid \tilde\epsilon_{\mathbf{k}} \le \tilde\mu\}$$

At half filling, the Fermi surface coincides with the boundary of the MBZ:

$$\delta = 0 \qquad\Longrightarrow\qquad \mathrm{FS} = \partial\mathrm{MBZ}$$

This is implied by the fact that renormalised bands are antisymmetric by $\boldsymbol{\pi}$ shifts,

$$\tilde\epsilon_{\mathbf{k}\sigma} = -\tilde\epsilon_{\mathbf{k}+\boldsymbol{\pi}\sigma}$$

The portion of an s^*-wave band outside the MBZ is mirrored with respect to the portion inside the MBZ. This implies that half the states lie in the MBZ, half out.

At $\delta>0$ the FS expands in the region $\mathrm{BZ}\setminus\mathrm{MBZ}$ where the form factor $\cos k_x+\cos k_y$ becomes negative. This gives the self-consistency equation

$$\lim_{\beta\to\infty} u^{(s^*)}[\beta,\tilde\mu] \simeq \frac{1}{L_xL_y}\sum_{\epsilon_{\mathbf{k}}\le 0}|\cos k_x+\cos k_y| - \frac{1}{L_xL_y}\sum_{0<\epsilon_{\mathbf{k}}\le\tilde\mu}|\cos k_x+\cos k_y|$$

which, being the sum of a positive and a negative term, is maximised whenever $\tilde\mu=0$, at half filling. This holds as long as the band is not flat. Indeed, if the effective hopping is not zero, to rise the doping level means to

$$\delta_1>\delta_2 \qquad\Longrightarrow\qquad \mathrm{Int}(\mathrm{FS}_1)\subset\mathrm{Int}(\mathrm{FS}_2)$$

with obvious meaning of the notation. Using this property we conclude

$$\tilde t>0 \quad\text{and}\quad \delta_1>\delta_2 \qquad\Longrightarrow\qquad \lim_{\beta\to\infty} u^{(s^*)}[\beta,\tilde\mu]\Big|_{\delta_1} < \lim_{\beta\to\infty} u^{(s^*)}[\beta,\tilde\mu]\Big|_{\delta_2}$$

This is the first expected behaviour of the self-consistent functional $u^{(s^*)}$: if there exists a region of parametric space where the shrinking effect is not strong enough to flatten the band, a rise in the doping level must lower $u^{(s^*)}$. Such a region must exist: at least all the segment $(V=0,\delta)$ is included, since on said region of parametric space the model is just the common free electronic model.

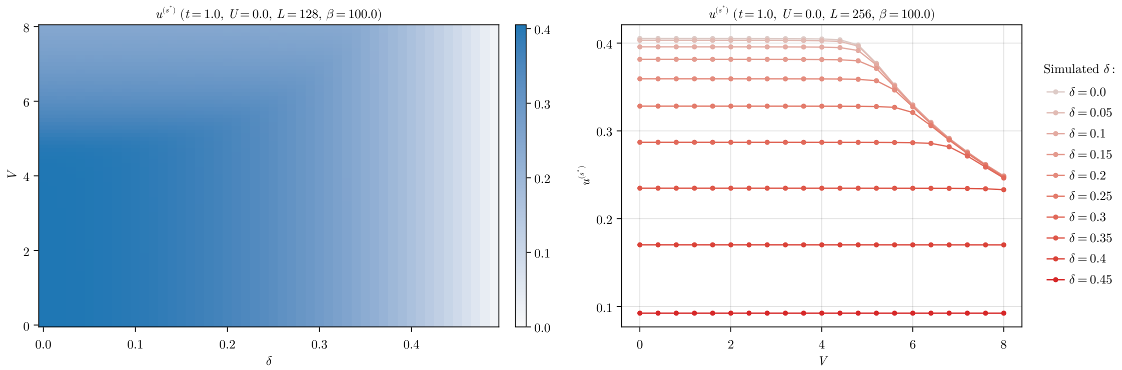
Following this line of reasoning, we can extract the two low-temperature limits of $u^{(s^*)}$ at half-filling and unitary filling.

1. At half-filling, the self-consistency equation reduces to

$$u^{(s^*)}[\beta,0]\big|_{\beta\gg1} \simeq \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}(\cos k_x+\cos k_y)$$

In thermodynamic limit, performing integration on the MBZ shown in the top-left panel of Fig. 1.3:

$$\begin{aligned}\lim_{\beta\to\infty} u^{(s^*)}[\beta,0] &\simeq \frac{1}{(2\pi)^2}\int_{-\pi}^{\pi}\mathrm{d}k_y\int_{-\pi+|k_y|}^{\pi-|k_y|}\mathrm{d}k_x\cos k_x + \frac{1}{(2\pi)^2}\int_{-\pi}^{\pi}\mathrm{d}k_x\int_{-\pi+|k_x|}^{\pi-|k_x|}\mathrm{d}k_y\cos k_y\\ &= \frac{1}{2\pi^2}\int_{-\pi}^{0}\mathrm{d}k_y\int_{-\pi-k_y}^{\pi+k_y}\mathrm{d}k_x\cos k_x + \frac{1}{2\pi^2}\int_{0}^{\pi}\mathrm{d}k_y\int_{-\pi+k_y}^{\pi-k_y}\mathrm{d}k_x\cos k_x\\ &= \frac{1}{\pi^2}\int_{-\pi}^{0}\mathrm{d}k_y\sin(\pi+k_y) + \frac{1}{\pi^2}\int_{0}^{\pi}\mathrm{d}k_y\sin(\pi-k_y)\\ &= \left(\frac{2}{\pi}\right)^2 \approx 0.405 \end{aligned} \tag{1.39}$$



a | Heatmap of $u^{(s^*)}$ in δV plane.

b | V-dependency of $u^{(s^*)}$ varying δ.

Figure 1.6 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase. To extract the data on the right side coarse-graining precision of the left-hand side data has been exchanged with higher lattice dimensions. As is evident, based on doping level $u^{(s^*)}$ exhibits non-trivial regions of V-independence.

2. At unitary filling we have

$$\lim_{\beta \to \infty} u^{(s^*)} \left[\beta, 2\tilde{t}^{(s^*)}\right] = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} (\cos k_x + \cos k_y) = 0$$

which vanishes due to the structure factor periodicity by $\boldsymbol{\pi}$.

To summarise: around $V = 0$, $u^{(s^*)}$ is expected to descend monotonically from $4/\pi^2$ to 0 at increasing doping.

1.6 Discussion of numeric results

In this section we present and discuss the numeric results, the physical relevance of the found phenomena and the boundaries of the algorithm. Let us start by sketching the algorithm used to determine self-consistently the mean-field structure of the normal phase in parameter space. For the normal phase, the HFPs "vector" defined in Sec. **??** has just one component:

$$\mathbf{v} = \left[u^{(s^*)}\right]$$

As before, we treat spin as a pure degeneracy. The only HFP is characterised by the first self-consistency equation of Tab. 1.2. The resulting model has two RMPs:

$$\tilde{t}_{ij} \equiv \left(t - \frac{u^{(s^*)} V}{2}\right) \varphi_{ij}^{(s^*)}$$
$$\tilde{\mu} \equiv \mu + n(2zV - U)$$

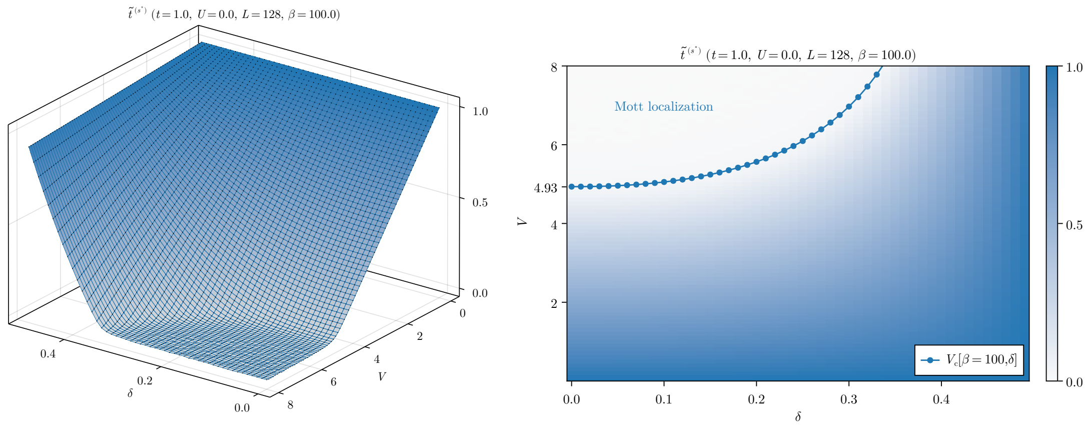
the second one being only dependent on the model parameters and not on HFPs. Finally, the phase free energy is given by Eq. (1.26). The normal phase is completely independent of U. Because of this we present the results varying the other parameters, keeping everywhere $U = 0$. We ran simulations in the (V, δ) plane, with the setup:

$$U = 0 \qquad \Delta n = 10^{-2} \qquad \Delta\mathbf{v} = \left[u^{(s^*)} : 5 \times 10^{-4}\right]$$

with $\Delta\mathbf{v}$ the tolerance vector defined as in Sec. **??**.

In Fig. 1.6a are reported the results obtained for the HFP $u^{(s^*)}$ on a 128×128 lattice and "low" temperature $\beta = 100$, varying the non-local attraction and the doping level

$$V = 0,\ 0.1,\ 0.2,\ \ldots,\ 4.0 \qquad \delta = 0,\ 0.01,\ 0.02,\ \ldots,\ 0.49$$



a | Simulated values of $\tilde{t}^{(s^*)}$

b | Critical Mott boundary.

Figure 1.7 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase and the Mott-boundary V_c, computed as is described in text. In the right panel, the tick $V = 4.93$ is specified explicitly as a graphic confirm of the computation of Eq. (1.40). For graphic reasons the points at $V_c > 8$ have been omitted.

The expectations of Sec. 1.5.3 are satisfied: at low V (indicatively $V \lesssim 5$) the trend is monotonically descendent at increasing doping. As V gets bigger, however, $u^{(s^*)}$ starts decreasing in the top-left corner of Fig. 1.6a. This effect is more visible in Fig. 1.6b, where we doubled L and ran simulations with a smaller δ resolution: the convergence value of $u^{(s^*)}$ is approximately independent of V up to some point, where this behaviour is broken. Note that, as was expected from the discussion of Sec. 1.5.3, the half-filled convergence value of $u^{(s^*)}$ in the V-independent region is approximately given by Eq. (1.39).

This effect can only be interpreted in one way, as we discuss in next section: $u^{(s^*)}$ maintains a constant value at fixed doping, as is described in Sec. 1.5.3, as long as $\tilde{t}^{(s^*)} > 0$. When V gets large enough, $u^{(s^*)}$ starts decreasing because of its constraints. The net effect is the nullification of $\tilde{t}^{(s^*)}$, which is, a total localisation effect.

1.6.1 Mott localisation effect

The most interesting result we obtained for the normal phase is the possibility of localising electrons by tuning the V/δ ratio. In Fig. 1.6b, at low V the convergence value of $u^{(s^*)}$ is approximately V-independent at sufficiently low temperatures. Let $V_c[\beta, \delta]$ be the critical value such that

$$0 \leq V < V_c[\beta, \delta] \quad : \quad u^{(s^*)} \text{ is } V \text{ independent.}$$

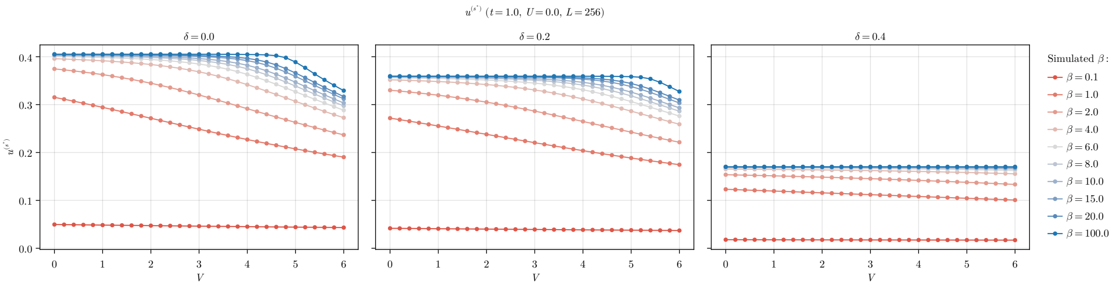
Then within the entire segment $[0, V_c]$ the convergence value is (approximately) the same obtained for $V = 0$. We can define in first instance the critical V as the value such that the hopping correction is maximized,

$$V_c[\beta, \delta] \equiv \frac{2t}{u^{(s^*)}[\beta, \tilde{\mu}(\delta)]\big|_{V=0}}$$

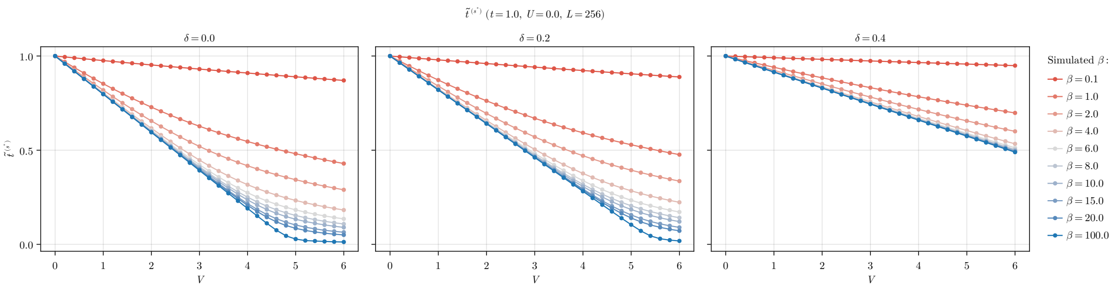
When V becomes larger than this critical value, the resulting RMP $\tilde{t}^{(s^*)}$ drops to 0 and due to the constraint of Eq. (1.33) the convergence value of $u^{(s^*)}$ is forced to decrease. Note that the constraint of Eq. (1.33) does not prevent $u^{(s^*)}$ from saturating to its upper boundary from below. In Fig. 1.7a the converged values of $\tilde{t}^{(s^*)}$, extracted from the same data of Fig. 1.6a, are plotted in parametric space.

To extract the value of $V_c[\beta, \delta]$ is a non-trivial analytic problem due to the insurgence of elliptic integrals and the smearing effects of temperature. Instead, for the high-β half-filled case we can use Eq. (1.39) and get:

$$V_c[\beta \to \infty, 0] = \frac{2t}{(2/\pi)^2} \approx 4.93t \tag{1.40}$$



a | Temperature dependence of $u^{(s^*)}$ at various dopings.



b | Temperature dependence of $\tilde{t}^{(s^*)}$ at various dopings.

Figure 1.8 | Comparative plots of the HFP $u^{(s^*)}$ (top panel) and RMP $\tilde{t}^{(s^*)}$ (bottom panel) in null-doped (left plots), "medium-doped" ($\delta = 0.2$, central plots) and "heavily-doped" ($\delta = 0.4$, right plots) contexts. Increasing temperature progressively destroys Mott localisation, reducing the bands shrinking effect.

At half-filling, the effective bands amplitude reduction is so strong that small values of V (compared to t) are sufficient to nullify kinetics. In Fig. 1.7b we report, superimposed, the convergence values of the RMP $\tilde{t}^{(s^*)}$ and $V_c[\beta = 100, \delta]$ computed with the simple procedure hereby described. Our prediction of the critical value for V approximates reasonably the localisation region.

Increasing temperature Mott localisation is suppressed. Also, out precedent argument were valid only at low temperature. In Fig. 1.8a we report results for the HFP $u^{(s^*)}$ computed at decreasingly lower temperature and three example doping levels,

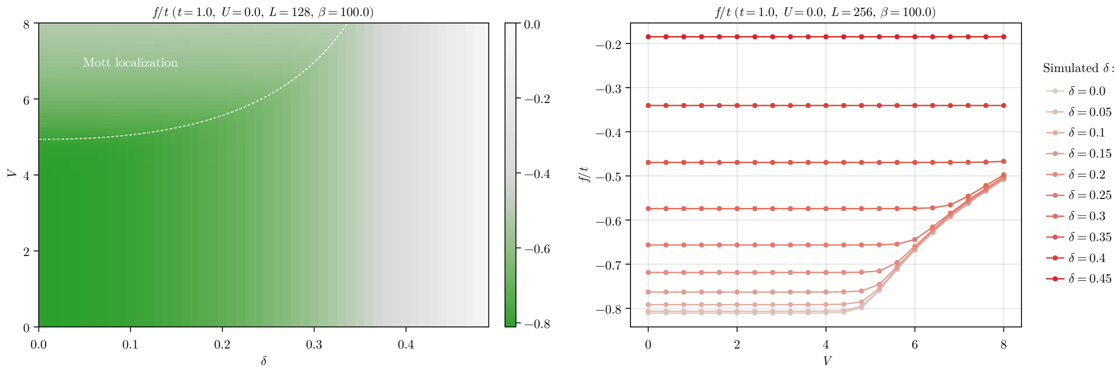
$$1.0 \leq \beta \leq 100.0 \qquad \delta = 0.0, 0.2, 0.4$$

At high temperature the V-independence region that was visible in Fig. 1.6b completely disappears, making $u^{(s^*)}$ a monotonically descendent function. In Fig. 1.8a the same dataset is plotted with focus on the RMP $\tilde{t}^{(s^*)}$. Temperature increase disrupts localisation effects, at least at small values of V. However, the bands shrinking effect is still relevant at $\beta = 1.0$, the hopping parameter approximately halving at $V \approx 5 \div 6$.

1.6.2 Solution free energy and entropy densities

In order to compare phases, we need an estimation of free energy of each. The plots of Fig. 1.9 report our findings on the normal phase free energy, computed based on Eq. (1.26). The free energy is intended in unities of t. For a non-flat band, doping increases energy because the band fills up and electrons populate higher energy levels.

Looking the data of Fig. 1.9a, at fixed δ free energy density remains pretty much constant up to the Mott boundary. This is simply because as long as a band with non-zero amplitude is present,



a | Heatmap of f in δV plane.

b | V-dependency of f varying δ.

Figure 1.9 | Free energy density of the normal phase. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 1.7b, indicating the Mott transition.

the HFP correction of Eq. (1.26) compensates the free energy increase due to bands shrinking,

$$
\begin{aligned}
-V\left|u^{(s^*)}\right|^2 &= -\frac{V u^{(s^*)*}}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\epsilon_{\mathbf{k}} - \tilde{\epsilon}_{\mathbf{k}})\, f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})
\end{aligned}
$$

having we used in the second passage

$$
\tilde{\epsilon}_{\mathbf{k}} = \tilde{\epsilon}_{\mathbf{k}} + u^{(s^*)*} V (\cos k_x + \cos k_y)
$$

Recalling the first point of the derivation of Sec. 1.4, to add the HFP correction to free energy is the same as removing the *tilde* sign from single-state energies, thus using the free model ($V = 0$) free energy density expression.

Mott localisation is energetically expensive. This is visible in both panels of Fig. 1.9. The V-dependent energy increase is an entropic effect: indeed, for a flat band:

$$
f(0; \beta, \tilde{\mu}(\delta)) = \frac{1}{2}
$$

at any temperature. The expression for entropy density of a free Fermi gas [29] is given by

$$
s(f) = -k_{\mathrm{B}} \left[f \log f + (1 - f) \log(1 - f) \right]
$$

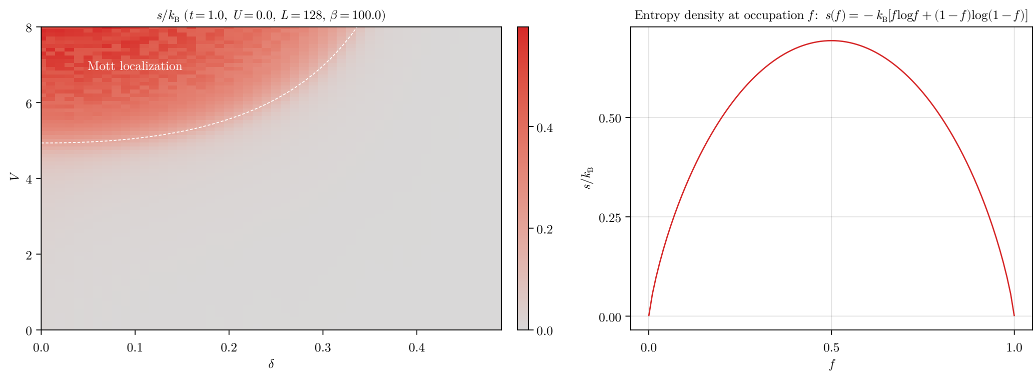
represented in Fig. 1.10b. The maximum is located at $f = 1/2$, then a flat band realises the maximally entropic condition. The Mott insulating region can be seen as a degeneration of the FS interface to the entire BZ, a process that maximises entropy.

All of this holds, as all the rest, at low temperature: by lowering β, finite-temperature effects become more and more relevant and the statements above need to be corrected.

1.6.3 Convergence problems and mixing fine-tuning

The normal phase is simple enough to describe some of the typical convergence problems arising in HF iterative schemes anticipated in Sec. **??**. Bands flattening is a source of algorithmic instability: numerics can lead to values of $u^{(s^*)}$, during iterations, slightly out of bounds with respect to Eq. (1.33). Even if the numeric effect is small, the consequence can be vast. To heal this effect and extract the actual convergence values, it is necessary to optimise the mixing parameter $g \in [0, 1]$.

We defined g in Sec. **??**, and operatively in Eq. (**??**). This parameter controls deterministically how, from an initialiser (the HFPs vector we plug in the algorithm) and an output (the HFPs



a | Heatmap of s_{MFT} in δV plane.

b | V-dependency of f varying δ.

Figure 1.10 | Entropy density of the normal phase (left panel) and theoretical contribution to entropy for a state with occupation f. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 1.7b, indicating the Mott transition.

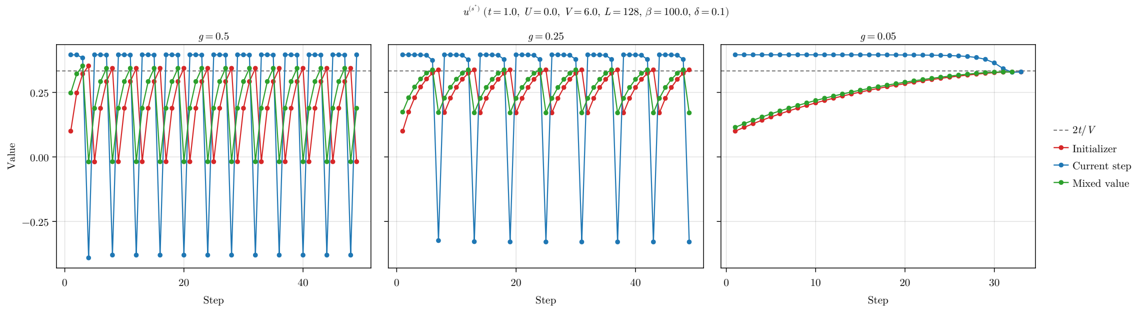


Figure 1.11 | Comparison of three example tracks for the single-valued HFPs vector of the normal phase, simulated at a "Mott-deep" point.

vector we get back from the algorithm) the new initialiser for the following step is generated. We show in Fig. 1.11 the evolutions at different mixing parameters of a simulation at

$$V = 6.0 \qquad \delta = 0.1 \qquad \beta = 100.0$$

This point is inside the Mott region, as can be checked in Fig. 1.6a. The HFP $u^{(s^*)}$ needs to converge at $2t/V = t/3$ in order to give complete localisation. The initialiser

$$\text{Step 1} \quad : \quad u^{(s^*)} = 0.1$$

was arbitrarily chosen. In each panel of Fig. 1.11, the red track represents the value of $u^{(s^*)}$ we plug in the self-consistency equations, the blue the result of the self-consistent equation, the green one the mixed value of the two based on different gs. The gray dashed line is the theoretical convergence target.

In the first two panels, there is an evident infinite loop occurring whenever the green line crosses the threshold $2t/V$. At those steps, the green point is out of bounds with respect to Eq. (1.33). When this happens, the following red point gives a band with inverted sign, a feature that "breaks" the self-consistency equation and generates a blue point with no physical significance.

To overcome this problem it is necessary to move "slowly". As is shown in the right panel, the choice of a small enough mixing parameter ensures convergence. In the track, each blue dot is above the target and would produce a band with inverted sign. Mixing blue and red dots with a 95% unbalance on the red ones, we ensure that the green track approaches the target from below. Note, however, that lowering g cannot be reiterated too many times. Indeed, for $g \to 0$, any point of our map becomes a fixed point.

This kind of problems can turn out to be pretty common for iHF schemes, and to understand how to knob the algorithmic parameters without sacrificing accurateness and precision, as well as avoiding computationally costly and frustrating infinite loops, was an essential part of this algorithm design.

Chapter 2

Superconducting instability

In this chapter the HF methods introduced in Sec. ?? are applied to the SC phase. We study translationally-invariant superconductivity. Analytic calculations will lead us to various effects: bare bands renormalisation, anisotropic Cooper pairing and nematicity. Our description of the SC phase is closely related to the BCS solution for pure s-wave superconductivity.

As for previous phases, this chapter is divided conceptually in three parts. First, analysis of the SC phase symmetries based on the discussion of Chap. ??. Second, implementation of HF methods and search for theoretic constraints on the resulting HFPs vector. Third, numeric results.

2.1 The SC phase and its symmetries

As for antiferromagnetism, superconductivity can establish in a variety of ways. We deal with the simplest realisation. In Sec. ?? we anticipated that the SC phase is described by the two SSBs

$$\mathrm{U}^z(1) \overset{\mathrm{SSB}}{\to} 0$$

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$$

Translational invariance is unbroken. Then electronic density needs to be uniform across all sites, similarly to the normal phase discussed in Chap. 1. Non-uniform density distributions are inconsistent with translational invariance in both directions. We have, then

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n \tag{2.1}$$

which is identical to Eq. (1.1) for the normal phase.

Recall Tabs. ?? and ??, collecting the selection rules for the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$. From Eqns. (??), (??) for the SC phase we need to account for the following contractions:

$\hat{H}_U$	Hartree and Cooper contractions
$\hat{H}_V$ (o.s. sector)	Hartree and Cooper contractions
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

Note that Cooper contractions in the non-local s.s. sector are neglected, because of the selection rule discussed at the end of Sec. ??. We anticipate the role of each contraction channel in Tab. 2.1.

In the SC phase we are breaking charge conservation $\mathrm{U}^c(1)$. This reflects in the fact that EVs like

$$\langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \rangle \neq 0 \qquad \text{(with } i,j \text{ NNs)}$$

are, in general, non-zero. The ground state plus two electrons can have a finite superposition with itself. This is a reformulation of the following statement: the HF ground-state wavefunction is written in terms of pairs of electrons bound together by some interaction. This is the phenomenon of Cooper pairing [28].

These pairs are composite quantum objects, with spatial and spin structure. Being couples of electrons, they must be antisymmetric under internal exchange. Moreover, as any composite

Operator	Sector	Channel	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
		Cooper	Isotropic pairing in singlet channel
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
		Cooper	Anisotropic pairing in singlet and triplet channels
	s.s.	Hartree	μ shift by nzV
		Fock	Band renormalisation and resymmetrisation

Table 2.1 Wick's contractions for each interaction and channel in the HF framework, and their net effect in the SC phase.

Spatial structure of Cooper pairs	Pairing channel
Inversion symmetric	Singlet pairing
Inversion anti-symmetric	Triplet pairing

Table 2.2 Relation between the spatial inversion (intended as $(x, y) \to (-x, -y)$) symmetry properties and the pairing channel for Cooper pairs.

object, a Cooper pair has a total spin whose algebra follows the composition rules of basic Quantum Mechanics,

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

which means that two $\mathfrak{su}(2)$ algebras are composed into a singlet and a triplet. The first is anti-symmetric under spin exchange; the second is symmetric. Hence, we can distinguish two Cooper pairing channels:

1. Singlet, with total spin projection equal to 0. In this channel, the Cooper pairs must exhibit spatial inversion symmetry.

2. Triplet, with total spin projection equal to 1. In this channel, the Cooper pairs must exhibit spatial inversion anti-symmetry.

Now, the HF hamiltonian will be expressed in terms of a set of HFPs which are linked to EVs of operators that create a Cooper pair. Hence, the symmetry properties of Cooper pairs reflects in those of the HF hamiltonian.

When dealing superconductivity, we can choose to restrict to just one of these pairing channels. In terms of the HF method, this means that we limit the variational search to the subset of states that are expressed in terms of pairs bound via a Cooper mechanism in the singlet or triplet channel. This limitations will provide specific selection rules on the superconducting gap function. In this thesis, we take on computations in general and in the end restrict to the singlet channel.

Let us rephrase: the spatial structure of Cooper pairs is linked to the composite spin state. This distinguishes two pairing channels, singlet and triplet. We separate these channels and limit search to just one. This operation will impose selection rules on the HFPs vector that are linked to the spatial symmetry of the gap function. This concept is summarised in Tab. 2.2.

We proceed as we did for the AF phase: in order to highlight the physical effects of $\hat{H}_V$, we first analyse the SC phase limitedly to the conventional HM. In next section we prove that BCS-like superconductivity is impossible at HF level on the pure HM, with just the local repulsion $\hat{H}_U$. It is the presence of the non-local attraction that enables Cooper pairing mechanism. Before starting, let

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{2.2}$$

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow}\hat{c}_{\mathbf{k}\uparrow} \tag{2.3}$$

where $\hat{n}_{\mathbf{k}\sigma}$ was defined as for the AF phase, in Eq. (**??**). The HF approximation in reciprocal space to the EHM will be written in terms of these operators.

2.2 (Impossibility of) superconductivity in the HM: application of HF methods

The conventional HM is given by $\hat{H}_t + \hat{H}_U$, thus its unique two-body interaction is the local repulsion. In this section we study the Cooper pairing channel in the conventional HM, and prove that superconductivity is impossible at HF level.

Following Tab. ?? and Eqns. (??), (??), we know the local repulsion decomposes in Hartree and Cooper channels,

$$\begin{aligned}\hat{H}_U &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow} \\ &\overset{\text{HF}}{\simeq} U \sum_{\mathbf{r}\in\mathcal{L}} \Big(\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\big\langle\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\big\rangle + \big\langle\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\big\rangle\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow} - \big\langle\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\big\rangle\big\langle\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\big\rangle && \text{(Hartree)} \\ &\qquad + \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\big\langle\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\big\rangle + \big\langle\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\big\rangle\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} - \big\langle\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\big\rangle\big\langle\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\big\rangle \Big) && \text{(Cooper)}\end{aligned}$$

In next sections we deal with each channel separately.

2.2.1 Local Hartree effects: chemical potential renormalisation

For the Hartree channel, being the density uniform across sites, we can re-use the entirety of Sec. ?? for the normal phase. Substituting Eq. (2.1), we get the approximation of Eq. (1.2),

$$\hat{H}_U \overset{\text{HF}}{\simeq} nU \times \hat{N} - E^{(\text{SC})}_{\text{H}/U} + (\text{Cooper channel}) \tag{2.4}$$

thus chemical potential renormalisation $\tilde{\mu} \to \mu - nU$ is present also in the SC phase. The contraction energy is the same of Eq. (??)

$$E^{(\text{SC})}_{\text{H}/U} = L_x L_y \times n^2 U \tag{2.5}$$

For a locally repulsive model, the Hartree channel contributes negatively to ground state energy. Indeed, contraction energy is subtracted from the hamiltonian. We now move to the Cooper channel.

2.2.2 Local Cooper channel effects: isotropic pairing

We move to reciprocal space, through the transformation of Eq. (??). Consider the Cooper channel of Eq. (??). In reciprocal space we get:

$$\begin{aligned}\hat{H}_U \overset{\text{HF}}{\simeq} (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \Big[&\big\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\big\rangle\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ &+ \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\big\langle\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}\big\rangle - \big\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\big\rangle\big\langle\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}\big\rangle\Big]\end{aligned} \tag{2.6}$$

We are not breaking translational invariance, thus only Cooper fluctuations with net zero total momentum are allowed. This means only $\mathbf{K} = \mathbf{0}$ contributes. Let us prove this:

Proof 2.1 (Cooper pairs have zero net momentum).

Consider the EV of the Cooper pair

$$\big\langle\hat{c}^\dagger_{\mathbf{k}_1\uparrow}\hat{c}^\dagger_{\mathbf{k}_2\downarrow}\big\rangle$$

for two momenta $\mathbf{k}_1$, $\mathbf{k}_2$. Moving to real space via an $\mathcal{F}$-transform, we get

$$\big\langle\hat{c}^\dagger_{\mathbf{k}_1\uparrow}\hat{c}^\dagger_{\mathbf{k}_2\downarrow}\big\rangle \overset{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{r}_1\in\mathcal{L}} \sum_{\mathbf{r}_2\in\mathcal{L}} e^{-i(\mathbf{k}_1\cdot\mathbf{r}_1 + \mathbf{k}_2\cdot\mathbf{r}_2)} \big\langle\hat{c}^\dagger_{\mathbf{r}_1\uparrow}\hat{c}^\dagger_{\mathbf{r}_2\downarrow}\big\rangle$$

where we used the convention of Eq. (??). Let now $\mathbf{r}_1 \to \mathbf{r}$ and $\mathbf{r}_2 \to \Delta\mathbf{r}$. Translational invariance implies that

$$g(\Delta\mathbf{r}) = \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\Delta\mathbf{r}\downarrow}\rangle$$

is independent of $\mathbf{r}$. Thus we have:

$$\begin{aligned}
\langle \hat{c}^\dagger_{\mathbf{k}_1\uparrow}\hat{c}^\dagger_{\mathbf{k}_2\downarrow}\rangle &\overset{\mathcal{F}}{=} \sum_{\Delta\mathbf{r}\in\mathcal{L}} e^{-i\mathbf{k}_2\cdot\Delta\mathbf{r}} g(\Delta\mathbf{r}) \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{-i(\mathbf{k}_1+\mathbf{k}_2)\cdot\mathbf{r}} \\
&= \sum_{\Delta\mathbf{r}\in\mathcal{L}} e^{-i\mathbf{k}_2\cdot\Delta\mathbf{r}} g(\Delta\mathbf{r}) \delta_{\mathbf{k}_1=\mathbf{k}_2}
\end{aligned}$$

where we used the identity of Eq. (**??**). Then the above EV vanishes for $\mathbf{k}_1 \neq \mathbf{k}_2$. Getting back to Eq. (2.6), this proves that only pairs at null total momentum can have nonzero EVs.

Now, let $w_\mathbf{k}$ be the EV of $\phi^\dagger_\mathbf{k}$, and $w^{(s)}$ its s-wave harmonic:

$$w_\mathbf{k} \equiv \langle \phi^\dagger_\mathbf{k} \rangle \tag{2.7}$$

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} w_\mathbf{k} \tag{2.8}$$

Eq. (2.6) reduces to

$$\hat{H}_U \overset{\mathrm{HF}}{\simeq} (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \left[\langle \hat{\phi}^\dagger_\mathbf{k} \rangle \hat{\phi}_{\mathbf{k}'} + \hat{\phi}^\dagger_\mathbf{k} \langle \hat{\phi}_{\mathbf{k}'} \rangle \right] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{2.9}$$

being $E^{(\mathrm{SC})}_{\mathrm{C}/U} = L_x L_y \times U \left| w^{(s)} \right|^2$ the HM Cooper contraction energy.

Proof 2.2 (Local Cooper contraction energy.).

The Cooper contraction energy for the local interaction is given by:

$$E^{(\mathrm{SC})}_{\mathrm{C}/U} \equiv \frac{U}{L_x L_y} \sum_{\mathbf{k}\mathbf{k}'} \langle \hat{\phi}^\dagger_\mathbf{k} \rangle \langle \hat{\phi}_{\mathbf{k}'} \rangle$$

Substituting Eqns. (2.7), (2.8) we get:

$$E^{(\mathrm{SC})}_{\mathrm{C}/U} = L_x L_y \times U \left(\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} w^*_\mathbf{k} \right) \left(\frac{1}{L_x L_y} \sum_{\mathbf{k}'\in\mathrm{BZ}} w_{\mathbf{k}'} \right) \tag{2.10}$$

$$= L_x L_y \times U \left| w^{(s)} \right|^2 \tag{2.11}$$

which proves our statement.

As for the Hartree channel, also the Cooper channel contributes negatively to ground-state energy. This could suggest that it is energetically favourable to develop s-wave superconductivity, but in the end is not.

We now define the s-wave component of the gap function,

$$\Delta^{(s)} \equiv -U w^{(s)} \tag{2.12}$$

Note that, in general, $\Delta^{(s)} \in \mathbb{C}$. However, we can always choose $\Delta^{(s)}$ as purely real. This is due to the fact that the hamiltonian is gauge-invariant, then it suffices to apply the gauge map of Eq. (**??**) to the fermionic operators with

$$\eta \equiv \frac{\arg \Delta^{(s)}}{2}$$

to obtain $\Delta^{(s)} \in \mathbb{R}$. For the sake of generality, we proceed with $\Delta^{(s)} \in \mathbb{C}$. Although this choice gives no advantage now, we implement it anyway in order to derive the most general form of the equations that follow. This is useful for the generalisation to the EHM.

In the end, we reduce $\hat{H}_U$ from Eq. (2.9) to

$$\hat{H}_U \overset{\text{HF}}{\simeq} (\text{Hartree channel}) - \sum_{\mathbf{k}\in\text{BZ}} \left[\Delta^{(s)}\hat{\phi}_{\mathbf{k}} + \Delta^{(s)*}\hat{\phi}^\dagger_{\mathbf{k}}\right] - E^{(\text{SC})}_{\text{C}/U} \tag{2.13}$$

This last decomposition resembles Eq. (**??**), obtained for the AF phase in the Hartree channel.

2.2.3 Superconductivity in the HM

We proceed here as we did for the AF phase in Sec. **??**. We start by giving a matrix representation of the approximated hamiltonian, then map it to a pseudospin problem and finally specify how to diagonalise it to retrieve its ground state.

Matrix representation. Collecting the results of the above paragraphs, we obtain the quadratic approximation to the Hubbard hamiltonian in the SC phase,

$$\begin{aligned}\hat{H}_t + \hat{H}_U \overset{\text{HF}}{\simeq} &\sum_{\mathbf{k}\in\text{BZ}}\sum_{\sigma} \epsilon_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \\ &- \sum_{\mathbf{k}\in\text{BZ}} \left[\Delta^{(s)}\hat{\phi}_{\mathbf{k}} + \Delta^{(s)*}\hat{\phi}^\dagger_{\mathbf{k}}\right] + nU \times \hat{N} - \left[E^{(\text{SC})}_{\text{H}/U} + E^{(\text{SC})}_{\text{C}/U}\right]\end{aligned} \tag{2.14}$$

We define the shifted bare bands:

$$\xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

Notice we are using the renormalised chemical potential. Due to inversion symmetry, it holds $\xi_{\mathbf{k}} = \xi_{-\mathbf{k}}$. Using Fermi anticommutation rules, the kinetic operator can be written as:

$$\begin{aligned}\hat{H}_t - \tilde{\mu}\hat{N} &= \sum_{\sigma}\sum_{\mathbf{k}\in\text{BZ}} \xi_{\mathbf{k}\sigma}\hat{n}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k}\in\text{BZ}} \left(\xi_{\mathbf{k}}\hat{n}_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}}\hat{n}_{-\mathbf{k}\downarrow}\right) \\ &= \sum_{\mathbf{k}\in\text{BZ}} \xi_{\mathbf{k}}\left(\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow} + 1\right) \\ &= \sum_{\mathbf{k}\in\text{BZ}} \xi_{\mathbf{k}}\left(\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow}\right) + E^{(\text{SC})}_{\text{a}}\end{aligned}$$

being $E^{(\text{SC})}_{\text{a}}$ the “anticommutation energy”

$$E^{(\text{SC})}_{\text{a}} \equiv \sum_{\mathbf{k}\in\text{BZ}} \xi_{\mathbf{k}} \tag{2.15}$$

In the end, the hamiltonian is approximated (in the sense of HF methods) by:

$$\begin{aligned}\hat{H}_t + \hat{H}_U \overset{\text{HF}}{\simeq} &\sum_{\mathbf{k}\in\text{BZ}} \epsilon_{\mathbf{k}}\left[\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{\mathbf{k}\downarrow}\hat{c}^\dagger_{\mathbf{k}\downarrow}\right] + E^{(\text{SC})}_{\text{a}} \\ &- \sum_{\mathbf{k}\in\text{BZ}} \left[\Delta^{(s)}\hat{\phi}_{\mathbf{k}} + \Delta^{(s)*}\hat{\phi}^\dagger_{\mathbf{k}}\right] + nU \times \hat{N} - \left[E^{(\text{SC})}_{\text{H}/U} + E^{(\text{SC})}_{\text{C}/U}\right]\end{aligned}$$

with $E^{(\text{SC})}_{\text{H}/U}$, $E^{(\text{SC})}_{\text{C}/U}$ the contraction energies arising from Hartree and Cooper channels, and $E^{(\text{SC})}_{\text{a}}$ the energy shift resulting from anticommutation rules, given respectively in Eqns. (2.5), (2.11), and (2.15). We recognised the s-wave gap $\Delta^{(s)}$ from Eq. (2.12).

Let the Nambu spinor

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^\dagger_{-\mathbf{k}\downarrow} \end{bmatrix}$$

EV	Meaning
$\langle \hat{s}^x_{\mathbf{k}} \rangle$	$\left\langle \hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}} \right\rangle$
$\langle \hat{s}^y_{\mathbf{k}} \rangle$	$i \left\langle \hat{\phi}_{\mathbf{k}} - \hat{\phi}^\dagger_{\mathbf{k}} \right\rangle$
$\langle \hat{s}^z_{\mathbf{k}} \rangle$	$\langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} \rangle - 1$

Table 2.3 List of the various expectation values (EVs) of each pseudospin component and their meaning in terms of fermionic operators.

The approximated hamiltonian can be written in matrix form as:

$$\hat{H}_t + \hat{H}_U - \mu \hat{N} \overset{\text{HF}}{\simeq} \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \hat{\Phi}^\dagger_{\mathbf{k}} h_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} + E^{(\text{SC})}_{\text{a}} - \left[E^{(\text{SC})}_{\text{H}/U} + E^{(\text{SC})}_{\text{C}/U} \right]$$

being

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta^*_{\mathbf{k}} \\ -\Delta_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix} \quad \text{with} \quad \xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

and $\Delta_{\mathbf{k}} = \Delta^{(s)}$.

Pseudospin representation. The hamiltonian can be written equivalently as

$$h_{\mathbf{k}} = \xi_{\mathbf{k}} \tau^z - \text{Re}\{\Delta_{\mathbf{k}}\} \tau^x - \text{Im}\{\Delta_{\mathbf{k}}\} \tau^y$$

Let now the pseudospin operator $\hat{\mathbf{s}}_{\mathbf{k}}$

$$\hat{s}^\alpha_{\mathbf{k}} \equiv \hat{\Phi}^\dagger_{\mathbf{k}} \tau^\alpha \hat{\Phi}_{\mathbf{k}} \quad \text{with} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{s}^x_{\mathbf{k}\sigma} \\ \hat{s}^y_{\mathbf{k}\sigma} \\ \hat{s}^z_{\mathbf{k}\sigma} \end{bmatrix}$$

and the pseudofield $\mathbf{b}_{\mathbf{k}}$:

$$\mathbf{b}_{\mathbf{k}} \equiv \begin{bmatrix} -\text{Re}\{\Delta_{\mathbf{k}}\} \\ -\text{Im}\{\Delta_{\mathbf{k}}\} \\ \xi_{\mathbf{k}} \end{bmatrix}$$

For a purely real gap, the pseudofield has no y component. In general, different gauges rotate the pseudofield around the z axis.

As was for the AF phase, these operators satisfy a $\mathfrak{su}(2)$ algebra up to an constant, irrelevant in the context of this discussion. They satisfy the identities:

$$\hat{s}^x_{\mathbf{k}} = \hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}} \tag{2.16}$$

$$\hat{s}^y_{\mathbf{k}} = i \left(\hat{\phi}_{\mathbf{k}} - \hat{\phi}^\dagger_{\mathbf{k}} \right) \tag{2.17}$$

$$\hat{s}^z_{\mathbf{k}} = \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow} - 1 \tag{2.18}$$

We report the EVs related to these identities Tab. 2.3.

In pseudospin representation, the approximated hamiltonian can be expressed as:

$$\hat{H}_t + \hat{H}_U - \mu \hat{N} \overset{\text{HF}}{\simeq} \sum_{\mathbf{k} \in \text{BZ}} \mathbf{b}_{\mathbf{k}} \cdot \hat{\mathbf{s}}_{\mathbf{k}} + nU\hat{N} + E^{(\text{SC})}_{\text{a}} - \left[E^{(\text{SC})}_{\text{H}/U} + E^{(\text{SC})}_{\text{C}/U} \right] \tag{2.19}$$

This operator is similar enough to the one found for the AF phase, in Eq. (**??**). It describes a set of pseudospin subject to a uniform field. As for the AF phase, we define the Bogoliubov bands $E_{\mathbf{k}}$

$$E_{\mathbf{k}} \equiv \sqrt{\xi^2_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^2}$$

The key difference with the AF Bogoliubov bands of Eq. (**??**) is that $E_{\mathbf{k}}$ now are functions of $\tilde{\mu}$, since $\xi_{\mathbf{k}}$ depends on the chemical potential. In other words, now the Bogoliubov bands depend on the shifted bands $\xi_{\mathbf{k}}$ instead of the bare bands $\epsilon_{\mathbf{k}}$. This means that the gap is opened at the chemical potential, not at zero energy, as was for the AF phase.

Moreover, let us define the angles $\theta_{\mathbf{k}}$, $\zeta_{\mathbf{k}}$ as

$$\cos 2\theta_{\mathbf{k}} = \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \qquad \sin 2\theta_{\mathbf{k}} \cos 2\zeta_{\mathbf{k}} = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \qquad \sin 2\theta_{\mathbf{k}} \sin 2\zeta_{\mathbf{k}} = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}$$

Note that choosing the gauge $\eta = \arg \Delta^{(s)}/2$ the imaginary part of the gap is zero, and this reflects in $2\zeta_{\mathbf{k}} = k \times 2\pi$ with $k \in \mathbb{Z}$.

Diagonalisation. For the SC phase, diagonalisation of the HF hamiltonian follows the same procedure as was for the AF phase. Indeed, the diagonalising matrix is given by

$$\begin{aligned} W_{\mathbf{k}} &\equiv e^{-i\theta_{\mathbf{k}}\tau^y} e^{-i\zeta_{\mathbf{k}}\tau^z} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} & -\sin\theta_{\mathbf{k}} \\ \sin\theta_{\mathbf{k}} & \cos\theta_{\mathbf{k}} \end{bmatrix} \begin{bmatrix} e^{-i\zeta_{\mathbf{k}}} & \\ & e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \end{aligned} \tag{2.20}$$

and the graphic procedure is the same of Fig. **??**, with $\epsilon \to \xi$. We get:

$$d_{\mathbf{k}} = W_{\mathbf{k}} h_{\mathbf{k}} W_{\mathbf{k}}^\dagger \quad \text{with} \quad d_{\mathbf{k}} \equiv \begin{bmatrix} E_{\mathbf{k}} & \\ & -E_{\mathbf{k}} \end{bmatrix} \tag{2.21}$$

In the tilted frame we have the expression for the Nambu spinors:

$$\hat{\Gamma}_{\mathbf{k}} = W_{\mathbf{k}} \hat{\Psi}_{\mathbf{k}} \tag{2.22}$$

Explicitly:

$$\begin{aligned} \hat{\Gamma}_{\mathbf{k}} &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^\dagger_{-\mathbf{k}\downarrow} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} - \sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} + \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}} \\ \hat{\gamma}^{(-)}_{\mathbf{k}} \end{bmatrix} \end{aligned} \tag{2.23}$$

The new Fermionic operators describe a Fermi gas on distributed on two mirrored bands,

$$\hat{H}_t + \hat{H}_U - \tilde{\mu}\hat{N} \overset{\mathrm{HF}}{\simeq} \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} E_{\mathbf{k}} \left(\hat{\gamma}^{(+)\dagger}_{\mathbf{k}} \hat{\gamma}^{(+)}_{\mathbf{k}} - \hat{\gamma}^{(-)\dagger}_{\mathbf{k}} \hat{\gamma}^{(-)}_{\mathbf{k}} \right) + (\cdots)$$

with $(\cdots)$ the various energy shifts.

2.2.4 Features of the SC solution

In this section we give some details about the HF solution for the translationally-invariant SC phase. We prove that in the conventional HM, with local repulsion $\hat{H}_U$, isotropic BCS-like superconductivity is forbidden. This serves as ground to motivate the passage to the EHM.

Pseudospin EVs. The EVs of pseudospin operators in the various directions are given by:

$$\begin{aligned} \langle \hat{\Psi}^\dagger_{\mathbf{k}} \tau^x \hat{\Psi}_{\mathbf{k}} \rangle &= -\sin 2\theta_{\mathbf{k}} \cos 2\zeta_{\mathbf{k}} \langle \hat{\Gamma}^\dagger_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \\ \langle \hat{\Psi}^\dagger_{\mathbf{k}} \tau^y \hat{\Psi}_{\mathbf{k}} \rangle &= -\sin 2\theta_{\mathbf{k}} \sin 2\zeta_{\mathbf{k}} \langle \hat{\Gamma}^\dagger_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \\ \langle \hat{\Psi}^\dagger_{\mathbf{k}} \tau^z \hat{\Psi}_{\mathbf{k}} \rangle &= \cos 2\theta_{\mathbf{k}} \langle \hat{\Gamma}^\dagger_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \end{aligned}$$

In the tilted frame, the averaged z projection of pseudospin is:

$$\langle \hat{\Gamma}^\dagger_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle = f\left(E_{\mathbf{k}}; \beta, 0\right) - f\left(-E_{\mathbf{k}}; \beta, 0\right) \tag{2.24}$$

Compared with the respective relation for the AF phase, Eq. (**??**), the main difference here is that the chemical potential inside the Fermi-Dirac distribution is set to zero. This is due to the fact that $\tilde{\mu}$ is already included in the renormalised bands $E_{\mathbf{k}}$. The approximated quadratic hamiltonian describes a gas of γ fermions at null chemical potential.

Let us define the thermal factors:

$$\mathrm{Th}_{\mathbf{k}}^{(\pm)}[\beta] \equiv f\left(-E_{\mathbf{k}};\beta,0\right) \pm f\left(E_{\mathbf{k}};\beta,0\right) \tag{2.25}$$

In the end we have the EVs:

$$\langle \hat{s}_{\mathbf{k}}^{x} \rangle = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \mathrm{Th}_{\mathbf{k}}^{(-)}[\beta] \tag{2.26}$$

$$\langle \hat{s}_{\mathbf{k}}^{y} \rangle = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \mathrm{Th}_{\mathbf{k}}^{(-)}[\beta] \tag{2.27}$$

$$\langle \hat{s}_{\mathbf{k}}^{z} \rangle = -\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \mathrm{Th}_{\mathbf{k}}^{(-)}[\beta] \tag{2.28}$$

Note that, as is expressed in Tab. 2.3, the z component of the pseudospin is related to density, being $1 + \langle \hat{s}_{\mathbf{k}}^{z} \rangle = \langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} \rangle$.

s-wave gap self-consistency equation. Applying the HF method to the conventional HM in SC phase, we obtained just one HFP: $w^{(s)}$. Let us derive its self-consistency equation. First, let

$$\tau^{\pm} \equiv \frac{\tau^{x} \pm i\tau^{y}}{2}$$

Recall Eq. (2.8) and expand it, to obtain

$$\begin{aligned} w^{(s)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{\phi}_{\mathbf{k}}^{\dagger} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{\Phi}_{\mathbf{k}}^{\dagger} \tau^{+} \hat{\Phi}_{\mathbf{k}} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\langle \hat{s}_{\mathbf{k}}^{x} \rangle + i \langle \hat{s}_{\mathbf{k}}^{y} \rangle}{2} \\ &= \frac{1}{2 L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\Delta_{\mathbf{k}}}{E_{\mathbf{k}}} \mathrm{Th}_{\mathbf{k}}^{(-)}[\beta] \end{aligned} \tag{2.29}$$

The last line is the explicit form of the self-consistency equation for $w^{(s)}$.

Impossibility of s-wave superconductivity. At HF level, the HM cannot host superconductivity. This means that $w^{(s)} = 0$. To show this, we need to define the critical U,

$$U_{\mathrm{c}}[\beta] \equiv \left[\frac{1}{2 L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\mathrm{Th}_{\mathbf{k}}^{(-)}[\beta]}{E_{\mathbf{k}}} \right]^{-1} \tag{2.30}$$

We call it "critical" because its value gives a sort of boundary for convergence. Indeed, **as we show in App. ??**, for the SC phase in conventional HM, we are able to optimise parametrically the mixing parameter g in order to ensure convergence, based on the ratio $U/U_{\mathrm{c}}[\beta]$. Let us show that superconductivity is not possible in the conventional HM with $U > 0$.

Proof 2.3 (Impossibility of s-wave superconductivity in the conventional HM).

By construction $U_c[\beta] > 0$, being the thermal factor positive by construction:

$$\begin{aligned}
\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] &= \frac{1}{e^{-\beta E_{\mathbf{k}}}+1} - \frac{1}{e^{\beta E_{\mathbf{k}}}+1} \\
&= \frac{e^{\beta E_{\mathbf{k}}}}{e^{\beta E_{\mathbf{k}}}+1} - \frac{1}{e^{\beta E_{\mathbf{k}}}+1} \\
&= \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right)
\end{aligned}$$

Using Eq. (2.30) Eq. (2.29) can be rewritten as:

$$w^{(s)} = -\frac{U}{U_c[\beta]} w^{(s)}$$

The unique possible self-consistent solution for the equation above is $w^{(s)} = 0$. This implication terminates the argument: superconductivity cannot establish self-consistently at HF level in the repulsive HM.

Note that the above proof does not hold for $U < 0$. An attractive model can host s-wave superconductivity.

Density. In order to estimate density, we can use the z projection of the pseudospin operators. From Eq. (2.18) we know

$$\hat{s}^z_{\mathbf{k}} + 1 = \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow}$$

Then, density can be estimated as:

$$\begin{aligned}
n &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow}\rangle \\
&= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} (\langle \hat{s}^z_{\mathbf{k}}\rangle + 1) \\
&= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \left(1 - \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]\right)
\end{aligned} \tag{2.31}$$

We can now move to the EHM.

2.3 Extension to the EHM

In this section we add to the hamiltonian the non-local attraction $\hat{H}_V$. As we did for the AF phase, we look for a solution with an identical mathematical structure as the one here described for the conventional HM. We define the renormalised quantities

$$\xi_{\mathbf{k}} \to \tilde{\xi}_{\mathbf{k}} \qquad E_{\mathbf{k}} \to \tilde{E}_{\mathbf{k}} \qquad \Delta_{\mathbf{k}} \to \tilde{\Delta}_{\mathbf{k}} \qquad \mu \to \tilde{\mu}$$

The chemical potential was already renormalised once in the HM derivation in Sec. **??**. Renormalisation of Bogoliubov bands follows that of bare bands and gap function,

$$\tilde{E}_{\mathbf{k}} \equiv \sqrt{\tilde{\xi}^2_{\mathbf{k}} + |\tilde{\Delta}_{\mathbf{k}}|^2}$$

Angles are changed too. Let $\tilde{\theta}$

$$\sin 2\tilde{\theta}_{\mathbf{k}} = \frac{|\tilde{\Delta}_{\mathbf{k}}|}{\tilde{E}_{\mathbf{k}}} \qquad \cos 2\tilde{\theta}_{\mathbf{k}} = \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tag{2.32}$$

and similarly the angle $\tilde{\zeta}_{\mathbf{k}}$,

$$\sin 2\tilde{\zeta}_{\mathbf{k}} = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \qquad \cos 2\tilde{\zeta}_{\mathbf{k}} = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \tag{2.33}$$

which give the pseudospin EVs

$$\langle\hat{\Phi}^\dagger_{\mathbf{k}}\tau^x\hat{\Phi}_{\mathbf{k}}\rangle = -\sin 2\tilde{\theta}_{\mathbf{k}}\sin 2\tilde{\zeta}_{\mathbf{k}}\langle\hat{\Gamma}^\dagger_{\mathbf{k}}\tau^z\hat{\Gamma}_{\mathbf{k}}\rangle \tag{2.34}$$

$$\langle\hat{\Phi}^\dagger_{\mathbf{k}}\tau^y\hat{\Phi}_{\mathbf{k}}\rangle = -\sin 2\tilde{\theta}_{\mathbf{k}}\cos 2\tilde{\zeta}_{\mathbf{k}}\langle\hat{\Gamma}^\dagger_{\mathbf{k}}\tau^z\hat{\Gamma}_{\mathbf{k}}\rangle \tag{2.35}$$

$$\langle\hat{\Phi}^\dagger_{\mathbf{k}}\tau^z\hat{\Phi}_{\mathbf{k}}\rangle = \cos 2\tilde{\theta}_{\mathbf{k}}\langle\hat{\Gamma}^\dagger_{\mathbf{k}}\tau^z\hat{\Gamma}_{\mathbf{k}}\rangle \tag{2.36}$$

Renormalisation of thermal factors and z pseudospin EV in tilted frame give:

$$\langle\hat{\Gamma}^\dagger_{\mathbf{k}}\tau^z\hat{\Gamma}_{\mathbf{k}}\rangle = f\left(\tilde{E}_{\mathbf{k}};\beta,0\right) - f\left(-\tilde{E}_{\mathbf{k}};\beta,0\right) \tag{2.37}$$

$$\tilde{\mathrm{Th}}^{(\pm)}_{\mathbf{k}}[\beta,\tilde{\mu}] \equiv f\left(-\tilde{E}_{\mathbf{k}};\beta,0\right) \pm f\left(\tilde{E}_{\mathbf{k}};\beta,0\right) \tag{2.38}$$

For the SC phase, the thermal factors are simpler than those of the AF phase. Indeed, using basic algebra, we see that:

$$\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}} = 1 \tag{2.39}$$

$$\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}} = \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.40}$$

Hence, pseudospin expectation values are:

$$\langle\hat{s}^x_{\mathbf{k}}\rangle = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.41}$$

$$\langle\hat{s}^y_{\mathbf{k}}\rangle = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.42}$$

$$\langle\hat{s}^z_{\mathbf{k}}\rangle = -\frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.43}$$

These equations are an extension of Eqns. (2.41), (2.42), (2.43) for the EHM. We can rewrite the equations for $w^{(s)}$ and n accordingly,

$$w^{(s)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.44}$$

$$n = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[1-\cos 2\tilde{\theta}_{\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right] \tag{2.45}$$

where we used Eqns. (2.29), (2.31).

We have set ground for the search of the HF solution for the EHM. From Tab. **??** we know the selection rules for Hartree, Fock and Cooper channel for the non-local attraction. We break down each.

2.3.1 Non-local Hartree effects: chemical potential renormalisation

We start by analysing the non-local Hartree channel,

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} -V\sum_{\langle ij\rangle}\sum_{\sigma,\sigma'}\left(\hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\langle\hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\rangle + \langle\hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\rangle\hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'} - \langle\hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\rangle\right)$$
$$+ (\text{Fock channel}) + (\text{Cooper channel})$$

where we used Eq. (**??**). This discussion is based on that for the normal phase in Sec. **??**, and leads to an identical result: the non-local Hartree channel renormalises the chemical potential.

We do not repeat the calculations already carried out for the normal phase. The non-local attraction $\hat{H}_V$ is reduced in the Hartree channel to

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} -2nzV \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/V} + (\text{all the rest})$$

This, together Eq. (2.4), gives the chemical potential full renormalisation, identical to Eq. (1.9),

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{2.46}$$

The contraction energy for the non-local sector was derived in Eq. (**??**),

$$E^{(\mathrm{SC})}_{\mathrm{H}/V} = -L_x L_y \times 2zn^2 V$$

Now we move to the more interesting Fock sector.

2.3.2 Fock effects: bare bands renormalisation

As done for the other phases, we here discuss the bands renormalisation effect derived from the Fock sector of the non-local attraction. From Eq. (**??**), we have

$$\begin{aligned}\hat{H}_V \overset{\mathrm{HF}}{\simeq}\ & (\text{Hartree channel}) \\ & V \sum_{\langle ij\rangle} \sum_{\sigma,\sigma'} \left(\hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} \rangle + \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \rangle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \rangle \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} \rangle \right) + (\text{Cooper channel})\end{aligned}$$

This discussion is essentially analogous the one done for the normal phase in Sec. 1.2.2. It will lead us to renormalisation of bare bands, which is given by a simultaneous shrinking of bare bands and emergence of an anisotropic component in hopping. As for previous phases, following the selection rules of Tab. **??**, only the s.s. sector contributes. Therefore we use from now on, in this section, $\sigma = \sigma'$.

In the context of the normal phase, when analysing the Fock channel, we got to Eq. (1.20). For the SC phase the same argument holds. Let us define, as done for the normal phase in Eq. (1.13) and for the AF phase in Eq. (**??**),

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle = \langle \hat{n}_{\mathbf{k}\sigma} \rangle$$

Hence, Eq. (1.20) becomes:

$$\begin{aligned}& V \sum_{\langle ij\rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \\ & \overset{\mathcal{F}}{=} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} u_{\mathbf{k}\sigma} \right] \times V \sum_{\mathbf{k}'\in \mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{k}'} \hat{c}^\dagger_{\mathbf{k}'\sigma} \hat{c}_{\mathbf{k}'\sigma} - E^{(\mathrm{SC})}_{\mathrm{F}/V} \end{aligned} \tag{2.47}$$

with the energy shift being given by:

$$E^{(\mathrm{SC})}_{\mathrm{F}/V} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle \langle \hat{c}^\dagger_{\mathbf{k}'\sigma} \hat{c}_{\mathbf{k}'\sigma} \rangle \tag{2.48}$$

$$= L_x L_y \times V \left[\left| u^{(s^*)} \right|^2 + \left| u^{(d)} \right|^2 \right] \tag{2.49}$$

where we used Eq. (**??**), which is formally identical in the SC phase with the only difference that EVs are now computed on the SC ground-state. To get to this point, we used the fact that:

$$\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma'} \rangle = \delta_{\mathbf{k}=\mathbf{k}'} \delta_{\sigma=\sigma'} \langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle$$

which states translational invariance and spin degeneracy.

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = \dfrac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) \left[1 - \dfrac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\dfrac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)\right]$	Fig. **??**
$d_{x^2-y^2}$-wave	$u^{(d)} = \dfrac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) \left[1 - \dfrac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\dfrac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)\right]$	Fig. **??**

Table 2.4 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$. The shown equations are reformulations of Eq. (2.51).

Let us define each γ-wave harmonic of $u_{\mathbf{k}}$ as

$$u_\sigma^{(\gamma)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle \tag{2.50}$$

The set of these harmonics give the HFPs in the Fock channel. The SC phase does not break inversion symmetry: then, as we concluded for the other phases, p_ℓ components are null. The renormalised bands are, as in Eqns. (1.21), (**??**),

$$\tilde{\epsilon}_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} + u^{(s^*)} V(\cos k_x + \cos k_y) + u^{(d)} V(\cos k_x - \cos k_y)$$

Each component satisfies the self-consistency equation:

$$u^{(\gamma)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \left[1 - \cos 2\tilde{\theta}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)\right] \tag{2.51}$$

In Tab. 2.4 we report these equations, writing explicitly the harmonics from Tab. **??** and using the angular relations of Eq. (2.32).

Let us prove the above equation. We proceed by proving first $u_{\mathbf{k}\uparrow} = u_{-\mathbf{k}\downarrow}$, and that both transform accordingly to Eq. (2.51)

Proof 2.4 (Self-consistency equations for $u^{(\gamma)}$).

We start by computing explicitly $u_{\mathbf{k}\sigma}$. For $\sigma = \uparrow$, we get the chain of equalities:

$$\begin{aligned} u_{\mathbf{k}\uparrow} &= \langle \hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{k}\uparrow} \rangle \\ &= \left\langle \hat{\Phi}_{\mathbf{k}}^\dagger \frac{\mathbb{I} + \tau^z}{2} \hat{\Phi}_{\mathbf{k}} \right\rangle \\ &= \frac{1}{2} \left[\left\langle \hat{\Gamma}_{\mathbf{k}}^\dagger \hat{\Gamma}_{\mathbf{k}} \right\rangle + \langle \hat{s}_{\mathbf{k}}^z \rangle \right] \\ &= \frac{1}{2} \left[\tilde{\mathrm{Th}}_{\mathbf{k}}^{(+)}[\beta] - \cos 2\tilde{\theta}_{\mathbf{k}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta] \right] \\ &= \frac{1}{2} \left[1 - \cos 2\tilde{\theta}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \right] \end{aligned} \tag{2.52}$$

having we used Eq. (2.38) and sequent. Inserting this result in Eq. (2.50), we get Eq. (2.51) in this spin sector. For $\sigma =\downarrow$,

$$\begin{aligned} u_{-\mathbf{k}\downarrow} &= \langle \hat{c}^\dagger_{-\mathbf{k}\downarrow}\hat{c}_{-\mathbf{k}\downarrow}\rangle \\ &= 1 - \langle \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow}\rangle \\ &= 1 - \left\langle \hat{\Phi}^\dagger_{\mathbf{k}} \frac{\mathbb{I}-\tau^z}{2} \hat{\Phi}_{\mathbf{k}} \right\rangle \\ &= 1 - \frac{1}{2}\left[\left\langle \hat{\Gamma}^\dagger_{\mathbf{k}}\hat{\Gamma}_{\mathbf{k}}\right\rangle - \langle \hat{s}^z_{\mathbf{k}}\rangle\right] \\ &= 1 - \frac{1}{2}\left[\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}}[\beta] + \cos 2\tilde{\theta}_{\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta]\right] \\ &= 1 - \frac{1}{2}\left[1 + \cos 2\tilde{\theta}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)\right] \end{aligned} \tag{2.53}$$

We observe that the last line equals Eq. (2.52). Hence

$$u_{\mathbf{k}\uparrow} = u_{-\mathbf{k}\downarrow}$$

which is coherent with our requirement of a uniform-density, spin uniform and inversion symmetric phase. This proves Eq. (2.51) for both spin sectors.

Note also that summing Eqns. (2.52), (2.53) we get the expression for density of Eq. (2.45). This is expected, since

$$n = \sum_{\mathbf{k}\in \mathrm{BZ}} \langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow}\rangle$$

Finally, if we take the $\tilde{\theta}_{\mathbf{k}} \to 0$ limit, we must retrieve the Fermi-Dirac distribution. Indeed, this limit coincides with $U, V \to 0$. Explicitly:

$$\begin{aligned} \lim_{\tilde{\theta}_{\mathbf{k}}\to 0} u_{\mathbf{k}\sigma} &= \frac{1}{2}\left[1 + \tanh\left(\frac{\beta\tilde{\xi}_{\mathbf{k}}}{2}\right)\right] \\ &= \frac{1}{2}\left[\frac{e^{\beta\tilde{\xi}_{\mathbf{k}}/2} + e^{-\beta\tilde{\xi}_{\mathbf{k}}/2}}{e^{\beta\tilde{\xi}_{\mathbf{k}}/2} + e^{-\beta\tilde{\xi}_{\mathbf{k}}/2}} - \frac{e^{\beta\tilde{\xi}_{\mathbf{k}}/2} - e^{-\beta\tilde{\xi}_{\mathbf{k}}/2}}{e^{\beta\tilde{\xi}_{\mathbf{k}}/2} + e^{-\beta\tilde{\xi}_{\mathbf{k}}/2}}\right] \\ &= f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

Correctly, the null-gap limit (for which the angle reduces to zero) gives back a perfectly degenerate Fermi gas. These considerations end the algebraic derivations for the Fock channel. We pass to the Cooper part.

2.3.3 Non-local Cooper effects: generalised pairing

In this section we discuss the effects of Cooper contractions in the HF decomposition of $\hat{H}_V$, shown in Eq. (**??**),

$$\begin{aligned} \hat{H}_V \overset{\mathrm{HF}}{\simeq} &\ (\text{Hartree channel}) + (\text{Fock channel}) \\ &- V \sum_{\langle ij\rangle}\sum_{\sigma,\sigma'} \left(\hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\big\langle \hat{c}_{j\sigma'}\hat{c}_{i\sigma}\big\rangle + \big\langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\big\rangle \hat{c}_{j\sigma'}\hat{c}_{i\sigma} - \big\langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\big\rangle\big\langle \hat{c}_{j\sigma'}\hat{c}_{i\sigma}\big\rangle\right) \end{aligned}$$

We know from Eq. (**??**) that the non-local attraction can be separated in o.s. and s.s. channels. As we reported in Tab. **??** and stated in Sec. 2.1, the only non-zero contribution comes from the o.s. sector. This sector contributes both to singlet and triplet pairing. Note that in singlet pairing channel the spatial wavefunction must be inversion-symmetric. On the contrary, in triplet pairing channel it needs to be inversion-antisymmetric.

Let us move to reciprocal space. From Eq. (**??**), we get

$$\hat{H}_V^{(\text{o.s.})} \overset{\text{HF}}{\simeq} (\text{Hartree channel}) - \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos\left(\delta k_x\right) + \cos\left(\delta k_y\right)\right] \times \Big[\left\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \right\rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \left\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \right\rangle - \left\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \right\rangle \left\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \right\rangle \Big]$$

Identical considerations as for the local repulsion hold: Cooper pairs have zero net momentum for translationally invariant SC phases. Hence, $\mathbf{K} = \mathbf{0}$. Therefore, the above operator is reduced to:

$$\hat{H}_V^{(\text{o.s.})} \overset{\text{HF}}{\simeq} (\text{Hartree channel}) - \frac{2V}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \left[\cos\left(\delta k_x\right) + \cos\left(\delta k_y\right)\right] \times \Big[\left\langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \right\rangle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} + \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \left\langle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \right\rangle - \left\langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \right\rangle \left\langle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \right\rangle \Big]$$

Recall now Eq. (2.7), where we defined

$$w_{\mathbf{k}} \equiv \langle \phi^\dagger_{\mathbf{k}} \rangle$$

Using the decomposition of Eq. (**??**), we end up with:

$$\hat{H}_V^{(\text{o.s.})} \overset{\text{HF}}{\simeq} (\text{Hartree channel}) - \frac{V}{L_x L_y} \sum_\gamma \sum_{\mathbf{k},\mathbf{k}'} \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}'} \left[w_{\mathbf{k}} \hat{\phi}_{\mathbf{k}'} + w^*_{\mathbf{k}} \hat{\phi}^\dagger_{\mathbf{k}'} \right] - E^{(\text{SC})}_{\text{C}/V/\text{o.s.}} \tag{2.54}$$

with the contraction energy being given by:

$$E^{(\text{SC})}_{\text{F}/V/\text{o.s.}} \equiv -\frac{V}{L_x L_y} \sum_\gamma \sum_{\mathbf{k},\mathbf{k}'} w_{\mathbf{k}} w^*_{\mathbf{k}'}$$

In both equations above, the sum is intended for $\gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$. The symmetry s is not included.

Now, the set of harmonics of $w_{\mathbf{k}}$ is part of the HFPs vector. In particular, as we will discuss in a moment, each define a different component of the superconducting gap function. The harmonics $w^{(\gamma)}$ are given by

$$w^{(\gamma)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} w_{\mathbf{k}} \tag{2.55}$$

This definition includes $w^{(s)}$. With this definition, we get the explicit form of the contraction energy:

$$E^{(\text{SC})}_{\text{F}/V/\text{o.s.}} = -L_x L_y \times V \sum_\gamma \left| w^{(\gamma)} \right|^2 \tag{2.56}$$

Note that the above sum does not contain $\gamma = s$. We now prove this.

Proof 2.5 (Non-local Cooper contraction energy.).

The contraction energy is formally given by:

$$E^{(\text{SC})}_{\text{C}/V/\text{o.s.}} \equiv -\frac{V}{L_x L_y} \sum_\gamma \sum_{\mathbf{k},\mathbf{k}'} \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}'} w_{\mathbf{k}} w^*_{\mathbf{k}'}$$

Symmetry	Self-consistency equation	Graph
s-wave	$w^{(s)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
s^*-wave	$w^{(s^*)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y)\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
p_x-wave	$w^{(p_x)} = \frac{i\sqrt{2}}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \sin k_x \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
p_y-wave	$w^{(p_y)} = \frac{i\sqrt{2}}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \sin k_y \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
$d_{x^2-y^2}$-wave	$w^{(d)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y)\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??

Table 2.5 | Symmetry resolved self-consistency equations for the harmonics of $w_{\mathbf{k}}$. The shown equations are reformulations of Eq. (2.57)

Substituting the harmonics decomposition of Eq. (2.55), we get:

$$\begin{aligned}
E^{(\mathrm{SC})}_{\mathrm{C/V/o.s.}} &= -L_xL_y \times V\sum_{\gamma}\left(\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}} w_{\mathbf{k}}\right)\left(\frac{1}{L_xL_y}\sum_{\mathbf{k}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{k}'} w_{\mathbf{k}'}\right)\\
&= -L_xL_y \times V\sum_{\gamma} w^{(\gamma)}w^{(\gamma)*}\\
&= -L_xL_y \times V\sum_{\gamma} \left|w^{(\gamma)}\right|^2 \qquad (\text{with } \gamma\in\{s^*, p_x, p_y, d_{x^2-y^2}\})
\end{aligned}$$

which proves our statement.

For a non-locally attractive model ($V > 0$) this contribution to energy is positive. This could suggest that the ground state would set $w^{(\gamma)} = 0$ in order to minimise energy. This is not the case, as we will see thoroughly in next sections. After all, the fact that this energy contribution is positive (which means, energetically unfavourable) does not mean anything without considering all contributions to energy.

Let us get back to the full operator. Applying the harmonic transformation of Eq. (2.55), Eq. (2.54) becomes

$$\hat{H}^{(\mathrm{o.s.})}_{V} \overset{\mathrm{HF}}{\simeq} (\text{Hartree channel}) - V\sum_{\gamma}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[w^{(\gamma)}\varphi^{(\gamma)*}_{\mathbf{k}}\hat{\phi}_{\mathbf{k}} + \text{h.c.}\right] - E^{(\mathrm{SC})}_{\mathrm{C/V/o.s.}}$$

Each harmonic, including s-wave, satisfies the self-consistency equation:

$$w^{(\gamma)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}}\sin 2\tilde{\theta}_{\mathbf{k}} e^{i2\tilde{\zeta}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.57}$$

In Tab. 2.5 we report these equations, writing explicitly the harmonics from Tab. **??** and using the angular relations of Eq. (2.32). Let us now give a proof to Eq. (2.57).

Proof 2.6 (Self-consistency equations for $w^{(\gamma)}$).

Consider Eq. (2.55). For the γ-wave component of the superconducting order parameter, the following chain, similar to Eq. (2.29), holds:

$$
\begin{aligned}
w^{(\gamma)} &\equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{\phi}_{\mathbf{k}}^{\dagger} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{\Phi}_{\mathbf{k}}^{\dagger} \tau^{+} \hat{\Phi}_{\mathbf{k}} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \frac{\langle \hat{s}_{\mathbf{k}}^{x} \rangle + i \langle \hat{s}_{\mathbf{k}}^{y} \rangle}{2} \\
&= \frac{1}{2 L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta]
\end{aligned}
$$

where $\tau^{+} = (\tau^{x} + i\tau^{y})/2$. We used Eqns. (2.34), (2.35) and (2.37). Use now Eqns. (2.32), (2.33),

$$
\begin{aligned}
\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} &= \frac{|\tilde{\Delta}_{\mathbf{k}}|}{\tilde{E}_{\mathbf{k}}} \left(\frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} + \frac{i\,\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \right) \\
&= \sin 2\tilde{\theta}_{\mathbf{k}} \left(\cos 2\tilde{\zeta}_{\mathbf{k}} + i \sin 2\tilde{\zeta}_{\mathbf{k}} \right) \\
&= \sin 2\tilde{\theta}_{\mathbf{k}} e^{i2\tilde{\zeta}_{\mathbf{k}}}
\end{aligned}
$$

and insert in previous equation. Finally, from Eq. (2.40), we can substitute the thermal factor with the hyperbolic tangent. This gives Eq. (2.57). The derivation above holds for all symmetries, (including s-wave).

With this proof we conclude this cumbersome algebraic derivation. In next section we summarise the obtained results, give the final form of the HF hamiltonian and discuss the symmetries of the gap function.

2.4 The HF hamiltonian and its properties

In this section we take together everything we said so far. Adding $\hat{H}_V$ to the conventional HM, we have seen, produces renormalisation of the bare bands, the gap function and the chemical potential.

By composing Hartree, Fock and Cooper channels we derive the final form of the HF gran-canonical hamiltonian,

$$
\hat{H} - \mu\hat{N} \overset{\mathrm{HF}}{\simeq} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\xi}_{\mathbf{k}} \left[\hat{c}_{\mathbf{k}\uparrow}^{\dagger} \hat{c}_{\mathbf{k}\uparrow} + \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{-\mathbf{k}\downarrow}^{\dagger} \right] - \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\tilde{\Delta}_{\mathbf{k}} \hat{\phi}_{\mathbf{k}} + \tilde{\Delta}_{\mathbf{k}}^{*} \hat{\phi}_{\mathbf{k}}^{\dagger} \right] + E_{\mathrm{a}}^{(\mathrm{SC})} + E_{\mathrm{c}}^{(\mathrm{SC})}
$$

having we included the "anticommutation energy" also defined for the conventional HM in Eq. (2.15),

$$
E_{\mathrm{a}}^{(\mathrm{SC})} \equiv \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\xi}_{\mathbf{k}}
$$

coming from fermionic anticommutation relations that are employed in order to get the kinetic term. We also defined the total contraction energy

$$
E_{\mathrm{c}}^{(\mathrm{SC})} \equiv -\left[E_{\mathrm{H}/U}^{(\mathrm{SC})} + E_{\mathrm{C}/U}^{(\mathrm{SC})} + E_{\mathrm{H}/V}^{(\mathrm{SC})} + E_{\mathrm{F}/V}^{(\mathrm{SC})} + E_{\mathrm{C}/V/\mathrm{o.s.}}^{(\mathrm{SC})} \right]
$$

which include all double counting terms we encountered. Note the minus sign in front of it.

2.4.1 Gap symmetries and and anisotropic pairing

We focus on the renormalised quantities. The bare bands are given by:

$$
\tilde{\xi}_{\mathbf{k}} = \epsilon_{\mathbf{k}} + V \sum_{\gamma\in\{s^*,d\}} u^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} - \tilde{\mu} \tag{2.58}
$$

and the full gap function by

$$\tilde{\Delta}_{\mathbf{k}} = -Uw^{(s)} + V\sum_{\gamma} w^{(\gamma)}\varphi_{\mathbf{k}}^{(\gamma)*} \tag{2.59}$$

We can identify the gap harmonics as

$$\tilde{\Delta}^{(s)} = -Uw^{(s)} \qquad \tilde{\Delta}^{(\gamma)} = Vw^{(\gamma)} \quad \text{being} \quad \gamma \in \{s^*, p_x, p_y, d\}$$

We will investigate competition among these components.

We anticipated in Tab. 2.2 that Cooper pairing in the singlet channel requires for the gap wavefunction to be inversion-symmetric. Triplet pairing, instead, requires inversion anti-symmetry. This gives a selection rule for the symmetries of the gap function: if we work in singlet channel, the gap must be symmetric, thus only three harmonics are allowed:

$$\text{Singlet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = \tilde{\Delta}^{(s)} + \tilde{\Delta}^{(s^*)}(\cos k_x + \cos k_y) + \tilde{\Delta}^{(d)}(\cos k_x - \cos k_y)$$

An inverse statement holds for the triplet pairing, for which

$$\text{Triplet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = i\sqrt{2}\left[\tilde{\Delta}^{(p_x)}\sin k_x + \tilde{\Delta}^{(p_y)}\sin k_y\right]$$

Before moving to the last part of this section, which gives an operative procedure to estimate the ground-state free energy and entropy densities, we dedicate a short section to the explicit form of the SC ground-state at zero temperature. We give a practical intepretation of the gap function as a parameter tuning the shape of the ground-state.

2.4.2 The zero-temperature ground-state at half-filling

The HF approximation to the true ground-state in the SC phase is obtained by aligning the each pseudospin with the respective pseudo-field. Let the up and down states,

$$|\Uparrow_{\mathbf{k}}\rangle \qquad |\Downarrow_{\mathbf{k}}\rangle$$

Using now Eq. (2.24), we have

$$\langle\Downarrow_{\mathbf{k}}|\hat{s}^z_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}}|\Downarrow_{\mathbf{k}}\rangle = 0$$
$$\langle\Uparrow_{\mathbf{k}}|\hat{s}^z_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}}|\Uparrow_{\mathbf{k}}\rangle = 2$$

so up to a phase

$$|\Downarrow_{\mathbf{k}}\rangle \sim |\Omega_{\mathbf{k}}\rangle \qquad |\Uparrow_{\mathbf{k}}\rangle \sim \hat{\phi}^\dagger_{\mathbf{k}}|\Omega_{\mathbf{k}}\rangle$$

The derivations carried out for the AF phase in Sec. **??** can be re-used. We end up with the state:

$$|\Psi\rangle = \bigotimes_{\mathbf{k}\in\mathrm{BZ}}\left[\cos\tilde{\theta}_{\mathbf{k}} + e^{2i\tilde{\zeta}_{\mathbf{k}}}\sin\tilde{\theta}_{\mathbf{k}}\hat{\phi}^\dagger_{\mathbf{k}}\right]|\Omega_{\mathbf{k}}\rangle$$

which is an extension of the conventional BCS state. The phase factor angle is the gap "tilt" in the complex plane. Indeed,

$$\begin{aligned}\langle\Psi|\hat{s}^x_{\mathbf{k}}|\Psi\rangle &= \langle\Psi|\hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}}|\Psi\rangle \\ &= \sin\tilde{\theta}_{\mathbf{k}}\cos\tilde{\theta}_{\mathbf{k}}\left(e^{2i\tilde{\zeta}_{\mathbf{k}}} + e^{-2i\tilde{\zeta}_{\mathbf{k}}}\right) \\ &= \sin 2\tilde{\theta}_{\mathbf{k}}\cos 2\tilde{\zeta}_{\mathbf{k}}\end{aligned}$$

$$\begin{aligned}\langle\Psi|\hat{s}^y_{\mathbf{k}}|\Psi\rangle &= i\,\langle\Psi|\hat{\phi}_{\mathbf{k}} - \hat{\phi}^\dagger_{\mathbf{k}}|\Psi\rangle \\ &= i\sin\tilde{\theta}_{\mathbf{k}}\cos\tilde{\theta}_{\mathbf{k}}\left(e^{-2i\tilde{\zeta}_{\mathbf{k}}} - e^{2i\tilde{\zeta}_{\mathbf{k}}}\right) \\ &= \sin 2\tilde{\theta}_{\mathbf{k}}\sin 2\tilde{\zeta}_{\mathbf{k}}\end{aligned}$$

These results are coherent with Eqns. (2.34) as well as (2.35) and allow us to identify the phase with angle of the pseudofield in the xy plane. The ground state out of half-filling, and at finite temperature, is in general more complicated but retains the same essential features. Although it is not trivial to write down an analytic expression for the state, we can extract its entropy and free energy densities quite straightforwardly. This is done in next sections.

2.4.3 The entropy density of the SC ground-state

The approximated ground-state in the SC phase is given by a gas of free $\hat{\gamma}^{(\pm)}$ fermions populating the two bands $\pm\tilde{E}_{\mathbf{k}}$ following a Fermi-Dirac distribution. Hence, the total entropy density of the ground state is given by that of a Fermi gas [29]:

$$s = s^{(-)} + s^{(+)}$$

where the superscript indicates the band,

$$s^{(\pm)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \Big[\Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\Big) \ln\Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\Big) \\ + f(\pm\tilde{E}_{\mathbf{k}};\beta,0) \ln f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\Big] \tag{2.60}$$

Note the fact that the Fermi-Dirac distributions are computed at null chemical potential. Because of this, we get:

$$s^{(-)} = s^{(+)} \qquad \Longrightarrow \qquad s \equiv s^{(-)} + s^{(+)} = 2s^{(+)}$$

Let us prove this.

Proof 2.7 (Bogoliubov bands $\pm\tilde{E}_{\mathbf{k}}$ give equal entropy densities.).

Consider the entropy of the lower band,

$$s^{(-)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \Big[\Big(1 - f(-\tilde{E}_{\mathbf{k}};\beta,0)\Big) \ln\Big(1 - f(-\tilde{E}_{\mathbf{k}};\beta,0)\Big) \\ + f(-\tilde{E}_{\mathbf{k}};\beta,0) \ln f(-\tilde{E}_{\mathbf{k}};\beta,0)\Big]$$

and consider the identity:

$$\frac{1}{e^{x}+1} + \frac{1}{e^{-x}+1} = 1$$

Which implies:

$$f(\epsilon;\beta,0) = 1 - f(-\epsilon;\beta,\mu)$$

Inserting the latter for $\epsilon = \tilde{E}_{\mathbf{k}}$ in the expression for $s^{(-)}$, we get

$$s^{(-)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \Big[f(\tilde{E}_{\mathbf{k}};\beta,0) \ln f(\tilde{E}_{\mathbf{k}};\beta,0) + \Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) \ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big)\Big]$$

hence $s^{(-)} = s^{(+)}$.

The two Bogoliubov bands give equal contribution to entropy. This is a consequence of the fact that for the SC phase the gap is always opened at the chemical potential, filling the lower band and symmetrically emptying the upper band.

The expression for the entropy density can be simplified by using Eq. [Missing], which implies

$$\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) \ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) = \ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) \\ - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big(\tilde{E}_{\mathbf{k}}\beta + \ln f(\tilde{E}_{\mathbf{k}};\beta,0)\Big)$$

It follows that $s = 2s^{(+)}$ can be written as:

$$s \equiv -\frac{2k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) + \frac{2k_B\beta}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} f(\tilde{E}_{\mathbf{k}};\beta,0)$$

Note that, as the gap closes and $\tilde{E}_{\mathbf{k}} \to \tilde{\xi}_{\mathbf{k}}$, the above expression reduces to the entropy density for the normal phase reported in Eq. [Missing].

2.4.4 The free energy density of the SC ground-state

Free energy density of the SC phase is derived following the procedure used for the other two phases. The five contributions to free energy are:

1. Quasiparticles energy:
$$\begin{aligned} e_{\mathrm{g}}^{(\mathrm{SC})} &= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta] \\ &= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \end{aligned}$$

2. Anticommutation energy:
$$\begin{aligned} e_{\mathrm{a}}^{(\mathrm{SC})} &\equiv \frac{E_{\mathrm{a}}^{(\mathrm{SC})}}{L_x L_y} \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\xi}_{\mathbf{k}} \end{aligned}$$

3. Contraction energy:
$$e_{\mathrm{c}}^{(\mathrm{SC})} = -U\left[n^2 + \left|w^{(s)}\right|^2\right] + V\left[2zn^2 - \sum_{\gamma}\left|u^{(\gamma)}\right|^2 + \sum_{\gamma}\left|w^{(\gamma)}\right|^2\right]$$

4. Entropic energy
$$\begin{aligned} e_{s}^{(\mathrm{SC})} &= -\frac{s}{k_{\mathrm{B}}\beta} \\ &= \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} f(\tilde{E}_{\mathbf{k}};\beta,0) \end{aligned}$$

5. Chemical potential correction
$$e_{n}^{(\mathrm{SC})} = \mu n$$

Now, compose everything
$$f^{(\mathrm{SC})} \equiv e_{\mathrm{g}}^{(\mathrm{SC})} + e_{\mathrm{a}}^{(\mathrm{SC})} + e_{\mathrm{c}}^{(\mathrm{SC})} + e_{s}^{(\mathrm{SC})} + e_{n}^{(\mathrm{SC})}$$

In particular, note that $e_{\mathrm{g}}^{(\mathrm{SC})}$ and the second term of $e_{s}^{(\mathrm{SC})}$ can be composed nicely,
$$-\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} f(\tilde{E}_{\mathbf{k}};\beta,0) = -\frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}}$$

Indeed, this is proven by noting that
$$\tanh\left(\frac{x}{2}\right) + \frac{2}{e^x + 1} = 1$$

which follows from basic algebra.

Taking everything together, in the end we get the following expression for free energy density:
$$\begin{aligned} f^{(\mathrm{SC})} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\tilde{\xi}_{\mathbf{k}} - \tilde{E}_{\mathbf{k}}\right] + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) \\ - U\left[n^2 + \left|w^{(s)}\right|^2\right] + V\left[2zn^2 - \sum_{\gamma}\left|u^{(\gamma)}\right|^2 + \sum_{\gamma}\left|w^{(\gamma)}\right|^2\right] + \mu n \end{aligned}$$

As was for the normal and AF phases, we recognise the chemical potential shift of Eq. (2.46),

$$f^{(\mathrm{SC})} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\tilde{\xi}_{\mathbf{k}} - \tilde{E}_{\mathbf{k}}\right] + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, 0)\right)$$
$$- U\left|w^{(s)}\right|^2 - V\left[\sum_{\gamma}\left|u^{(\gamma)}\right|^2 - \sum_{\gamma}\left|w^{(\gamma)}\right|^2\right] + \tilde{\mu} n \quad (2.61)$$

which gives the analytic expression for the free energy. Note that, as the gap closes and $\tilde{E}_{\mathbf{k}} \to \tilde{\xi}_{\mathbf{k}}$, the above expression reduces to the entropy density for the normal phase reported in Eq. [Missing].

2.4.5 Energy constraints on the HFPs

We derived the expression of Eq. (2.61), which contains in its second row the HFPs $u^{(\gamma)}$ and $w^{(\gamma)}$. In this section, similarly to what was done in Sec. [Missing]for the AF phase, we give a proof of the following equation:

$$-U\left|w^{(s)}\right|^2 + V\sum_{\gamma}\left|w^{(\gamma)}\right|^2 = \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \quad (2.62)$$

which we will use to derive a constraint on the HFPs themselves.

Proof 2.8 (Proof of Eq. (2.62)).

Recall the explicit form of the conjugate gap function from Eq. (2.59),

$$\tilde{\Delta}^*_{\mathbf{k}} = -U w^{(s)*} + V \sum_{\gamma} w^{(\gamma)*} \varphi^{(\gamma)}_{\mathbf{k}}$$

Expand the right-hand side of Eq. (2.62) and insert the above equation,

$$\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) = \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\Delta}^*_{\mathbf{k}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$$
$$= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[-U w^{(s)*} + V \sum_{\gamma} w^{(\gamma)*} \varphi^{(\gamma)}_{\mathbf{k}}\right] \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$$

In last line, the sum over γ is intended as limited to the symmetries s^*, p_x, p_y, d. Now, use the self-consistency equations of Tab. 2.5 to recognise the various harmonics. We reduce the right-hand side of Eq. (2.62) to:

$$\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) = -U w^{(s)*} \times \overbrace{\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)}^{w^{(s)}}$$
$$+ V \sum_{\gamma} w^{(\gamma)*} \times \underbrace{\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)}_{w^{(\gamma)}}$$

which coincides with the left-hand side.

The right-hand side of Eq.(2.62) is a sum of positive terms, hence is positive (or zero, if the gap is null). So must be the right hand side. Then we have the constraint

$$-U\left|w^{(s)}\right|^2 + V\sum_{\gamma}\left|w^{(\gamma)}\right|^2 \geq 0$$

If the gap is null, the above constraint is satisfied because $w^{(\gamma)} = 0$ for all γs. If the gap is finite, the above requirement translates to:

$$\sum_{\gamma} \left|w^{(\gamma)}\right|^2 \geq M^{(\mathrm{SC})} \quad \text{and} \quad \gamma \neq s \tag{2.63}$$

where we defined the radius:

$$M^{(\mathrm{SC})} \equiv \left|w^{(s)}\right|^2 \frac{U}{V}$$

Now, as we stated in Sec. 2.2.4, for the pure HM we have identically $w^{(s)} = 0$. For the EHM, instead, we see that:

- There is no reason *a priori* for $w^{(s)} = 0$ to be null;
- If $w^{(s)} \neq 0$ (in the singlet channel) the other harmonics – namely, s^* and d-wave – must "compensate" and be large enough, as is stated in Eq. (2.63). This means that $w^{(s)}$ cannot be non-zero without at least one of the other two singlet harmonics being non-zero;
- In the strong coupling limit $U/t \gg V/t$, the value of $M^{(\mathrm{SC})}$ can become rather large even for relatively small values of $w^{(s)}$. This is a convoluted way of saying that increasing U/t at fixed V/t we should expect the harmonic $w^{(s)}$ to decrease or the other two to increase (or a mix of these two behaviours).

In the singlet channel, adding the non-local attraction to the conventional HM fixes (in principle) the impossibility for superconductivity at HF level.

These considerations end the theoretical discussion of the SC phase. Unfortunately, we are not able to provide any more constraints on the HFPs, differently from AF and normal phase. Fortunately, as we will show with the analysis of numerics in next section, there is no particular need for additional constraints, at least for the $w^{(\gamma)}$ HFPs. We would have liked to have a proof that $0 \leq u^{(s^*)} \leq 2t/V$ also for the SC phase, but we do not have any.

2.5 Discussion of numeric results for the singlet channel

The last part of this chapter is devoted to presenting and discussing the numeric results we extracted from the iHF procedure. **We limited simulations to the singlet channel, because** [I wanted to see the end of this project!]. We specify that, differently from AF and normal phases, for the singlet SC phase we did not implement any theoretical constraint on the five HFPs,

$$\mathbf{v} = \begin{bmatrix} u^{(s^*)} \\ u^{(d)} \\ w^{(s)} \\ w^{(s^*)} \\ w^{(d)} \end{bmatrix} \qquad \mathbf{v} = \mathbb{R}^5$$

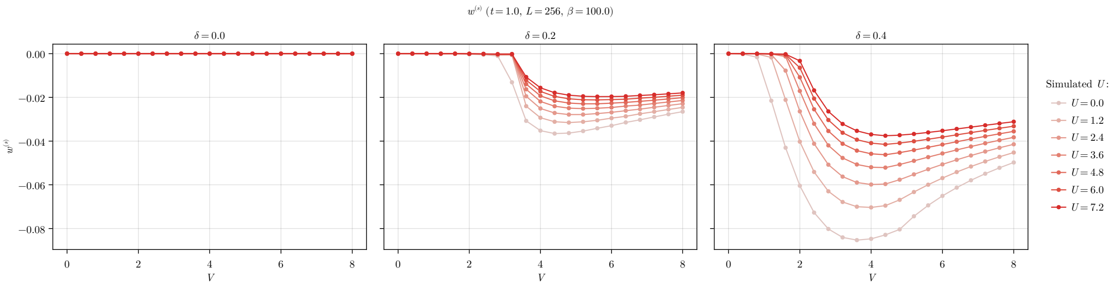
As we will discuss, all five HFPs are significantly non-zero in this phase, at least in some portion of parameters space. Therefore, we could at most search theoretical constraints describing the boundaries of these portions of space. We did not, and just ran unconstrained simulation by brute-force.

The results we obtained indicate that this phase is described by three RMPs:

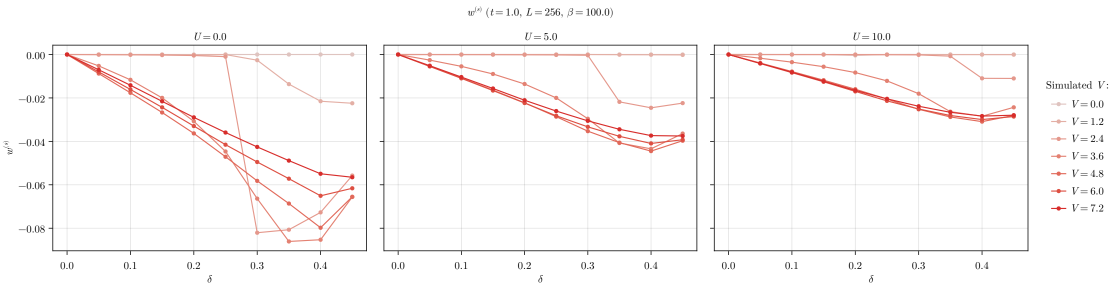
$$\tilde{t}_{ij}^{(s^*)} \equiv \left(t - \frac{u^{(s^*)}V}{2}\right)\varphi_{ij}^{(s^*)}$$
$$\tilde{t}_{ij}^{(d)} \equiv -\frac{u^{(s^*)}V}{2}\varphi_{ij}^{(d)}$$
$$\tilde{\mu} \equiv \mu + n(2zV - U)$$

For AF and normal phase, we had two HFPs and no d-wave component of the renormalised hopping. We ran simulations at $\beta = 100.0$ within the boundaries:

$$0 \leq U \leq 12 \qquad 0 \leq V \leq 8 \qquad 0 \leq \delta \leq 0.45$$



a | Dependency of $w^{(s)}$ on V at various Us, compared at three values of δ.



b | Dependency of $w^{(s)}$ on δ at various Vs, compared at three values of U.

Figure 2.1 | Dependency of the s-wave HFP $w^{(s)}$ on V at fixed U and δ (top panel, compared figures) and on δ at fixed V and U (bottom panel, compared figures).

For the normal phase we did not vary U; for the AF phase we did not vary δ. Here, we need to vary both in order to explore the entire parametric space.

2.5.1 Multi-symmetry singlet superconductivity

In singlet channel, the most general gap function is a combination of s, s^* and d components. The resulting gap function $\tilde{\Delta}_{\mathbf{k}}$ emerges from a competition among these channels. Running unconstrained simulations, we are able to see directly the result of the competition in the self-consistent solution. We discuss the three channels separately.

***s*-wave gap.** Consider the isotropic harmonic of the gap function, namely the s-wave component. In Fig. 2.1, we show the data for $w^{(s)}$. It is given by:

$$\tilde{\Delta}^{(s)} = -U w^{(s)}$$

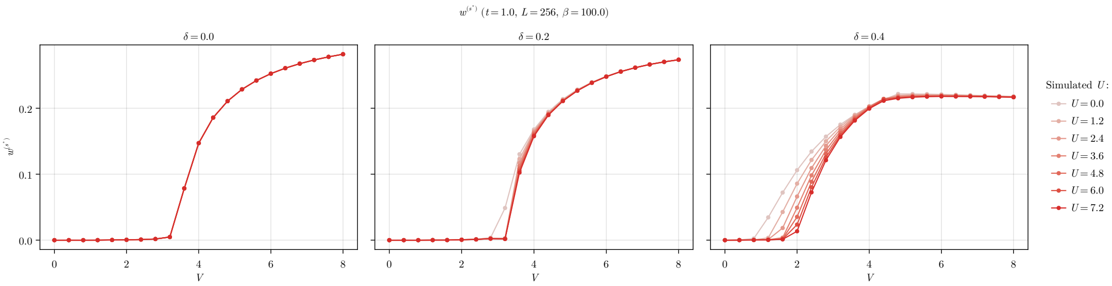
In the conventional HM we know that s-wave superconductivity is impossible at HF level. This was discussed in Sec. (2.2.4). Instead in the EHM there is no selection rule for $w^{(s)}$. The only constraint comes from Eq. (2.63): $w^{(s)}$ can be non-zero as long as at least one $w^{(\gamma)}$ with $\gamma = s^*, d$ is non-zero. This is a necessary but insufficient condition.

In Fig. 2.1a we see that $w^{(s)}$ is null at $\delta = 0$ (left panel). Hence

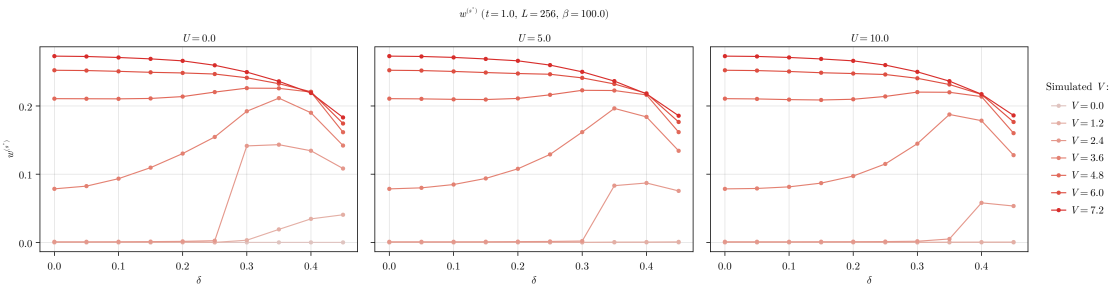
$$\forall \; U/t, V/t \quad : \quad \tilde{\Delta}^{(s)}\big|_{\delta=0} = 0 \tag{2.64}$$

at least within the range of parameters we explored. Increasing doping (central and right panels), $w^{(s)}$ is null up to a critical value $V_c^{(s)}$, which depends also on U. For $V \geq V_c^{(s)}$, $w^{(s)}$ is negative and the shape of the curve depends strongly on U. In particular, the amplitude $\left|w^{(s)}\right|$ exhibits a local maximum. At almost unitary filling, ratios V/t comparable with 1 are sufficient to develop a finite convergence value for $w^{(s)}$. Therefore, doping lowers the critical value of the non-local attraction that is needed in order to establish an isotropic gap in the local channel.

The dependence of $w^{(s)}$ on doping is shown in Fig. 2.1b. As anticipated, $w^{(s)} = 0$ at half-filling. Varying U/t and V/t we see that $w^{(s)} = 0$ reaches some negative minimum whose absolute value is lower at higher Us.



a | Dependency of $w^{(s^*)}$ on V at various Us, compared at three values of δ.



b | Dependency of $w^{(s^*)}$ on δ at various Vs, compared at three values of U.

Figure 2.2 | Dependency of the s^*-wave HFP $w^{(s^*)}$ on V at fixed U and δ (top panel, compared figures) and on δ at fixed V and U (bottom panel, compared figures).

s^*-wave gap. The results for the non-local isotropic HFP $w^{(s^*)}$ are shown in Fig. 2.2. This component of the full gap is given by

$$\tilde{\Delta}^{(s^*)} = V w^{(s^*)}$$

Consider first Fig. 2.2a. At null and finite fillings, $w^{(s^*)}$ is null up to a critical value $V_c^{(s^*)}$. Its position depends on δ and weakly on U/t. The half-filling curve (left panel) is independent of U/t. Differently from $w^{(s)}$, we have

$$\forall\ U/t \quad \wedge \quad \forall V < V_c^{(s^*)} \quad : \quad \tilde{\Delta}^{(s^*)}\big|_{\delta=0} = 0$$

The local isotropic gap harmonic $\tilde{\Delta}^{(s)}$ was always null at half filling. Its non-local counterpart $\tilde{\Delta}^{(s^*)}$ is null up to the critical value delimiting the phase transition. Therefore, superconductivity can establish at half-filling with a gap whose isotropic part is purely non-local. Note that we are not saying that pure s^*-wave superconductivity is possible. As we discuss later, at half-filling the dominant harmonic of the gap function has d-wave symmetry.

In Fig. 2.2b we see that the dependence on doping of $w^{(s^*)}$ is peculiar. At low doping ($\delta \lesssim 0.3$), $w^{(s^*)}$ is small for small values of V (left panel). At higher doping (central and right panels) we see that a local maximum emerges, indicating that $w^{(s^*)}$ converges to higher values at smaller Vs.

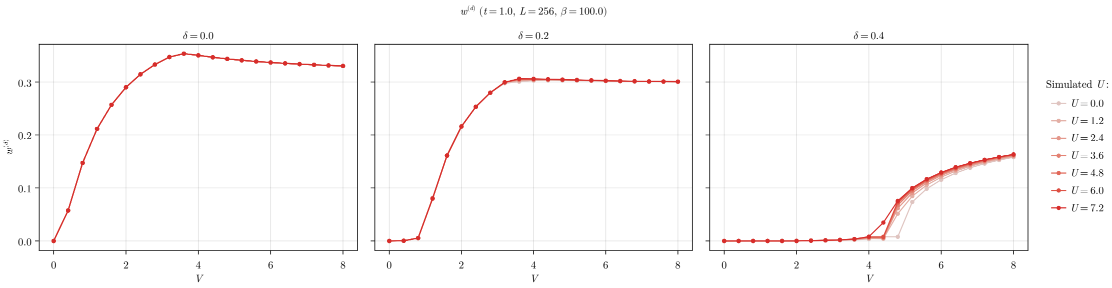
d-wave gap. In Fig. 2.3 we report our findings for the HFP $w^{(d)}$, recalling that the d-wave gap is given by:

$$\tilde{\Delta}^{(d)} = V w^{(d)}$$

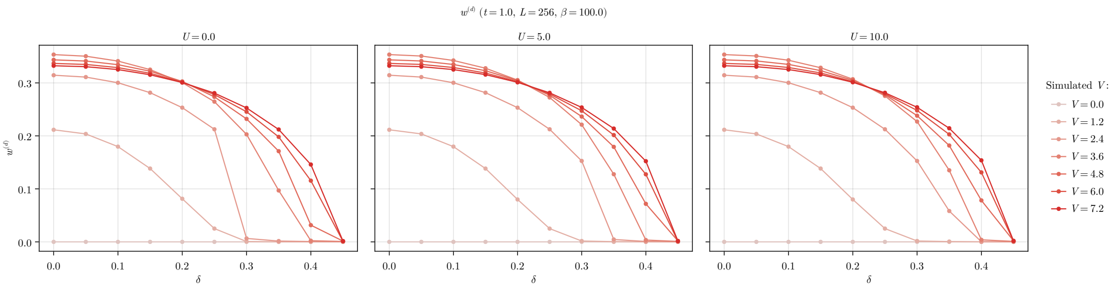
We see from the top panel (Fig. 2.3a) that $w^{(d)}$ depends weakly on U. At half-filling (left panel), the d-wave gap is zero only at $V = 0$:

$$\forall\ U/t \quad \wedge \quad V = 0 \quad : \quad \tilde{\Delta}^{(d)}\big|_{\delta=0} = 0$$

Differently from the other components, $w^{(d)}$ grows to finite values already at infinitesimal V/t. Recalling what we said in the above paragraph, we note that when the strength of the non-local



a | Dependency of $w^{(d)}$ on V at various Us, compared at three values of δ.



b | Dependency of $w^{(d)}$ on δ at various Vs, compared at three values of U.

Figure 2.3 | Dependency of the d-wave HFP $w^{(d)}$ on V at fixed U and δ (top panel, compared figures) and on δ at fixed V and U (bottom panel, compared figures).

attraction grows enough, the s^*-wave part sets in and symmetries mix. At finite filling (central and right panels) a critical value $V_c^{(d)}$ emerges such that the d-wave gap appears at $V > V_c^{(d)}$.

The dependence of $w^{(d)}$ on doping at different values of U is shown in the bottom row (Fig. 2.3b). For any V/t, the HFP $w^{(d)}$ is a monotonically decreasing function of δ. As we said, the critical value $V_c^{(d)}$ increases with doping. Because of this, we see that the lighter curves of Fig. 2.3b, corresponding to lower values of V, reach zero quite rapidly. The curves for larger values of V, instead, reach zero at almost unitary filling. In all three panels of Fig. 2.3b we see that at doping $\delta \approx 0.3$ various lines cross at $w^{(d)} \approx 0.3$. Finally, at null doping, there exists a finite value of V that maximises $w^{(d)}$: it is visible in Fig. 2.3a.

Competition among the channels. To get a valid understanding of the competition among channels, we need to compare the values of the harmonics $\tilde{\Delta}^{(\gamma)}$ for $\gamma = s, s^*, d$. Competition is reflected in the relative ratio between different components. We report a comparison of such in Fig. 2.4. **The presented data are limited to the case $U/t = 4$ for graphic reasons,but data and plots for other values of U are openly accessible at our repository.**

On the left column we plot three heatmaps, one for each harmonic in singlet SC phase. The same data are shown, superimposed, in the surface plot on the right. The harmonics $\tilde{\Delta}^{(\gamma)}$ cross if V/t gets large enough, with the d-wave component reaching the largest values in the parametric region we investigated, followed closely by the s^*-wave harmonic. Isotropic harmonics (s and s^*) are significantly non-zero in the same region, as we see in the first two rows of the left column. The two channel differ in amplitude significantly: the maximum value reached by the local gap harmonic $\tilde{\Delta}^{(s)}$ is approximately an order of magnitude smaller than the maximum of the non-local harmonic $\tilde{\Delta}^{(s^*)}$. Moreover, we see that at $\delta = 0.0$ the local isotropic harmonic $\tilde{\Delta}^{(s)}$ is completely suppressed, as we expressed in Eq. (2.64).

All harmonics share a quite strange boundary around $\delta \approx 0.34$, $V/t \approx 2.5$, visible as a sharp vertical segment in the three heatmaps on the left column of Fig. 2.4. The isotropic harmonics suddenly change from zero to large values moving from left to right, while the d-wave part does the opposite. Whether this behaviour underlies some deeper structure we were not able to tackle with our coarse-grained simulations, or is a numeric artifact, we do not know.

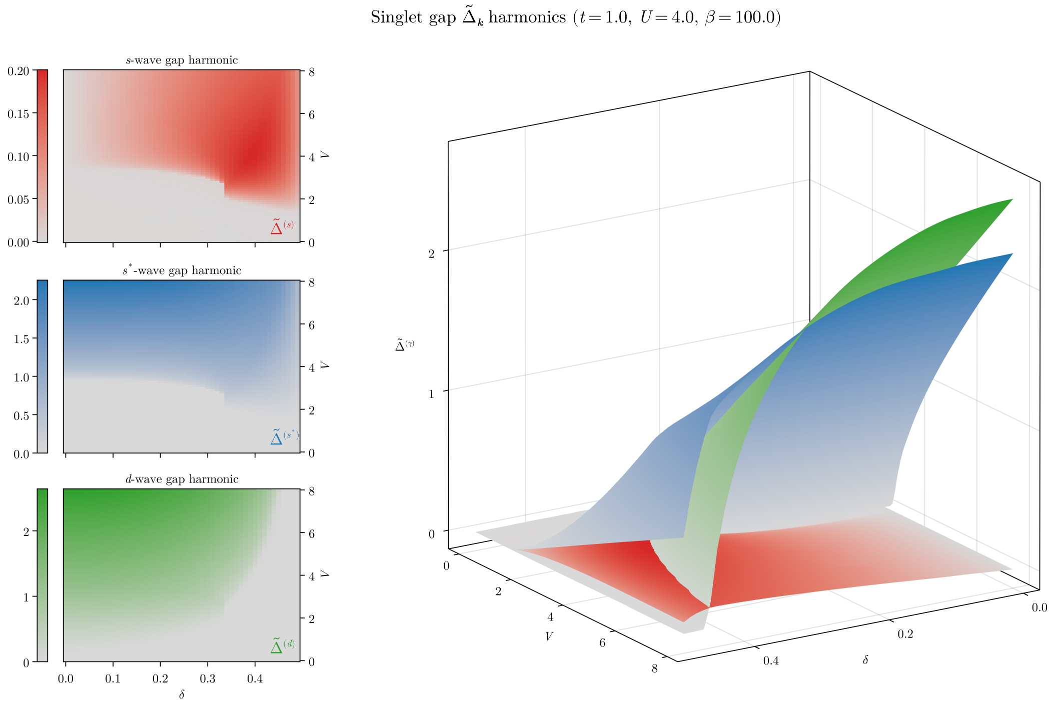


Figure 2.4 Plot of the singlet gap harmonics $\tilde{\Delta}^{(\gamma)}$, with $\gamma = s, s^*, d$ at $U/t = 4$. The panels on the left show the three harmonics separately as heatmaps, while the surface plot on the right shows them comparatively.

Four phases can be distinguished: the normal state and three superconducting regions (pure d-wave, pure $s \oplus s^*$-wave, mixed). In Fig. 2.5 we give a schematic depiction of how the plane (δ, V) divides in the four regions at $U/t = 0$ (left), $U/t = 4$ (centre) and $U/t = 12$ (right). The boundaries were computed by setting an arbitrary threshold $\varepsilon = 10^{-2}$, with the following rule:

$$\begin{array}{rcccc}
\text{Normal} & \left|\tilde{\Delta}^{(s)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(s^*)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(d)}\right| < \varepsilon \\
d\text{-wave} & \left|\tilde{\Delta}^{(s)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(s^*)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(d)}\right| \geq \varepsilon \\
s \oplus s^*\text{-wave} & \left[\left|\tilde{\Delta}^{(s)}\right| \geq \varepsilon \right. & \vee & \left. \left|\tilde{\Delta}^{(s^*)}\right| \geq \varepsilon\right] & \wedge & \left|\tilde{\Delta}^{(d)}\right| < \varepsilon \\
\text{Mixed} & \left[\left|\tilde{\Delta}^{(s)}\right| \geq \varepsilon \right. & \vee & \left. \left|\tilde{\Delta}^{(s^*)}\right| \geq \varepsilon\right] & \wedge & \left|\tilde{\Delta}^{(d)}\right| \geq \varepsilon
\end{array}$$

We consider s and s^* as part of the same isotropic channel: just one harmonic being larger than the threshold is sufficient for us to assert that the gap function has an isotropic harmonic.

Comparing the three panels of Fig. 2.5, we see that increasing the value of the local repulsion U the shapes of the four regions change. In particular, the d-wave region expands to the right (larger δs); the normal-$s \oplus s^*$ SC boundary moves up (larger Vs). Note that on the left panel the isotropic region is labeled as purely s^*-wave: this is because if $U = 0$, the s-wave gap harmonic is null, even if $w^{(s)}$ converges to a non-zero value. The growth of the local repulsion U has two effects:

1. turning on the s-wave gap harmonic;
2. expanding the d-wave region and restricting the isotropic and mixed region.

From the last point we understand that increasing U we allow for the gap function to have a finite s-wave component, but this acts as a sort of internal competition within the isotropic region. Indeed, from the constraint of Eq. (2.63), we know that $w^{(s)} \neq 0$ only if $w^{(\gamma)} \neq 0$, with $\gamma \neq s$. In the $s \oplus s^*$ region the constraint reads:

$$\left|w^{(s^*)}\right|^2 \geq \left|w^{(s)}\right|^2 \frac{U}{V} \qquad \text{(Isotropic region)}$$

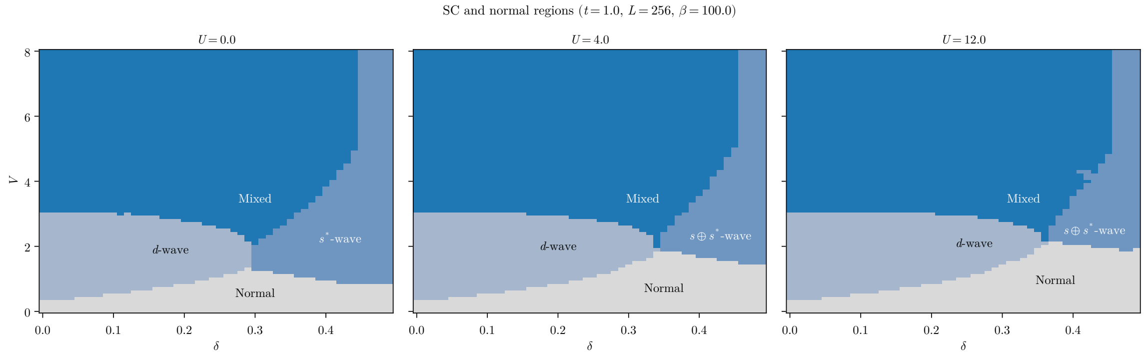


Figure 2.5 Depiction of the normal state region and the three singlet superconducting regions, at $U/t = 0, 4, 12$ from left to right. Lighter blue indicates the pure d-wave SC phase, blue the isotropic $s \oplus s^*$ phase, deeper blue the mixed region. In text is described the convention we used to identify the boundaries.

Suppose we work at fixed δ and $V > 0$ in the isotropic region at $U = 0$. There is no constraint whatsoever on $w^{(s^*)}$. Also, $w^{(s)}$ is free to converge at some finite value (we saw this in the left panel of Fig. 2.1b): the gap still has no s-wave component. Suppose now we switch on $U > 0$. Now, in order for $w^{(s)}$ to reach the same value as before, $w^{(s^*)}$ has to respect the constraint posed by the above equation. As U increases, the right-hand side gets bigger and $w^{(s^*)}$ needs to converge to larger and larger values to sustain $w^{(s)}$, otherwise to drop to zero. This is why increasing U the isotropic boundary moves up: $w^{(s^*)}$ isn't "free" anymore and must respect the constraint of Eq. (2.63).

In the three panels of Fig. 2.5, the mixing region is pretty much unchanged. At low doping, it is bounded from below by an horizontal line at $V/t \approx 3$. Increasing doping at fixed U, we see that the lower boundary reaches a cusp-like minimum around $\delta \approx 0.3 \div 0.4$ and then grows. This cusp coincides with the aforementioned sharp vertical segment, visible in Fig. 2.4. If we work at fixed doping and U, and choose δ such that we are aligned with the horizontal position of the cusp on the (δ, V) plane, the critical value of V/t for transitioning to a mixed SC state is minimised. At high values of V/t the mixing region extends to almost unitary doping, being limited on the right by a vertical line at $\delta \gtrsim 0.4$. We remind that we are discerning the SC phases by applying an arbitrary threshold ε to each gap component: we imagine that by using a more refined technique and by increasing the resolution of the simulation the boundary would appear less straight. Anyway, apart from a vertical stripe at almost unitary filling, at high V/t the dominant singlet SC phase is described by a mix of d-wave and $s \oplus s^*$-wave component. Recalling the surface plot of Fig. 2.4, we see that at $V/t \gg 1$ the s-wave component gets weaker with respect to the other two. In this limit, if $U \ll V$, the s-wave component is negligible, giving an approximate $d \oplus s^*$ mixed phase.

The mixing region is of particular interest also due its connection with the planar rotational SSB,

$$\text{Symmetry mixing} \quad \Longrightarrow \quad \mathrm{C}_4 \overset{\text{SSB}}{\to} \mathrm{C}_2$$

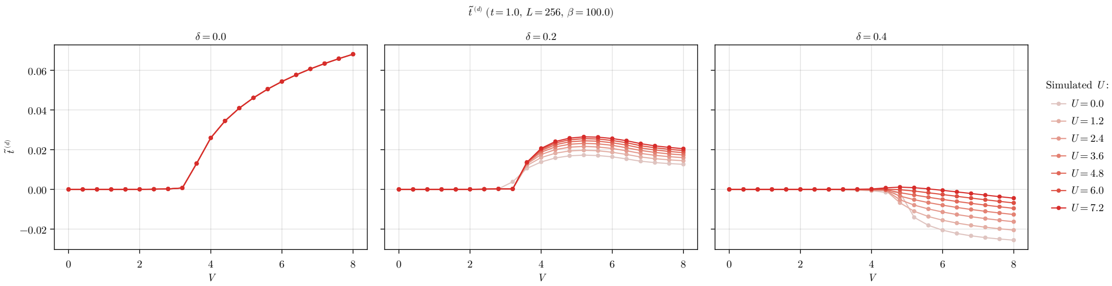
As we show in next section, the mixed-symmetry region coincides with the portion of the (δ, V) plane where planar rotational symmetry is broken.

2.5.2 Rotational SSB and nematic superconductivity

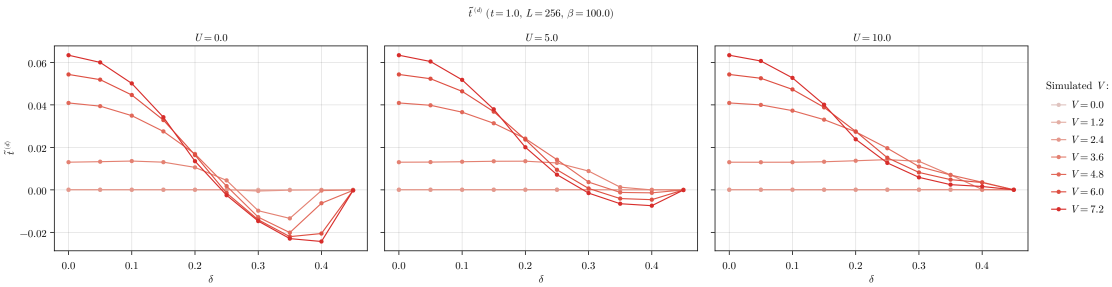
In the context of this thesis, nematicity is established whenever $\tilde{t}^{(d)}$ becomes non-zero. This was discussed in detail in the context of the normal phase. If the renormalised hopping parameter has a non-zero d-wave component, this means that bare hopping is effectively boosted in one direction and damped in the other. Indeed, from Eq. (1.24), we know that

$$\tilde{t}^{(d)} = \frac{\tilde{t}^{(y)} - \tilde{t}^{(x)}}{2}$$

As we anticipated at the end of Sec. 1.3.1, if $\tilde{t}^{(d)} > 0$ the renormalised hopping is larger in direction y, if $\tilde{t}^{(d)} < 0$ instead it's larger in direction x. This interpretation serves to grasp the general idea



a | Dependency of $\tilde{t}^{(d)}$ on V at various Us, compared at three values of δ.



b | Dependency of $\tilde{t}^{(d)}$ on δ at various Vs, compared at three values of U.

Figure 2.6 | Dependency of the d-wave renormalised hopping $t^{(d)}$ varying the doping δ and the local repulsion U. In the top panel we compare three doping levels $\delta = 0.0,\ 0.2,\ 0.4$ varying $0 \leq U \leq 8$. In the bottom panel we compare three repulsion values $U = 0.0,\ 5.0,\ 10.0$ varying $0 \leq \delta \leq 0.45$.

behind nematicity, but is to be taken carefully. Let us get more rigorous. In the SC phase, the many body state is a gas of Bogoliubov quasiparticles which do not “hop on neighbouring sites”. If $\tilde{t}^{(d)}$ is non-zero, the superconductor is nematic in the sense that Bogoliubov quasibands are not rotationally invariant,

$$\tilde{E}_{\mathbf{k}} \neq \tilde{E}_{\mathcal{R}\mathbf{k}}$$

being $\mathcal{R}$ a $\pi/2$ rotation. We connect this effect to hopping renormalisation *a posteriori* following the formal discussion of Sec. 1.3.1.

For the normal and AF phases we were able to prove that $u^{(d)} = 0$ for any choice of model parameters. For the SC phase, instead, $u^{(d)}$ converges to non-zero values. For the sake of conciseness, we skip plotting the curves for $u^{(d)}$ **(which are, anyway, openly accessible at our repository)**, and show directly the curves for the quantity

$$\tilde{t}^{(d)} = -\frac{V u^{(d)}}{2}$$

This is done in Fig. 2.6: nematicity is undoubtedly present. In the top panel (Fig. 2.6a), we see that a critical value of V exists over which nematicity is established. Moreover, the $\delta = 0.0$ case (left panel) is independent of U. In all cases, the convergence values for $\tilde{t}^{(d)}$ are two order of magnitudes smaller than $t = 1.0$: in the region of parametric space we explore, nematicity is a sharp but small effect. Note that at $\delta = 0.2$ (central panel) $\tilde{t}^{(d)} \geq 0$, while at $\delta = 0.4$ (right panel) $\tilde{t}^{(d)} \leq 0$. Increasing the doping we change the sign of $\tilde{t}^{(d)}$

This sign change is most evident in the first two figures of the bottom panel (Fig. 2.6b), where we vary the doping level while keeping fixed the local repulsion U. The behaviour of $\tilde{t}^{(d)}$ is similar at low doping in the three cases, but changes significantly at larger doping. Indeed, at $U/t = 0$ (left figure) and $U/t = 5$ (central figure) at some point $\tilde{t}^{(d)}$ changes sign. At $U/t = 10$ (right figure), all our data points are positive (possibly the sign flip occurs at higher Vs). Moreover, as U increases, the amplitude of the oscillation of $\tilde{t}^{(d)}$ as a function of δ reduces. We conclude that both the local repulsion and electronic doping disrupt nematicity, which is instead most effective for a purely NNs attractive model.

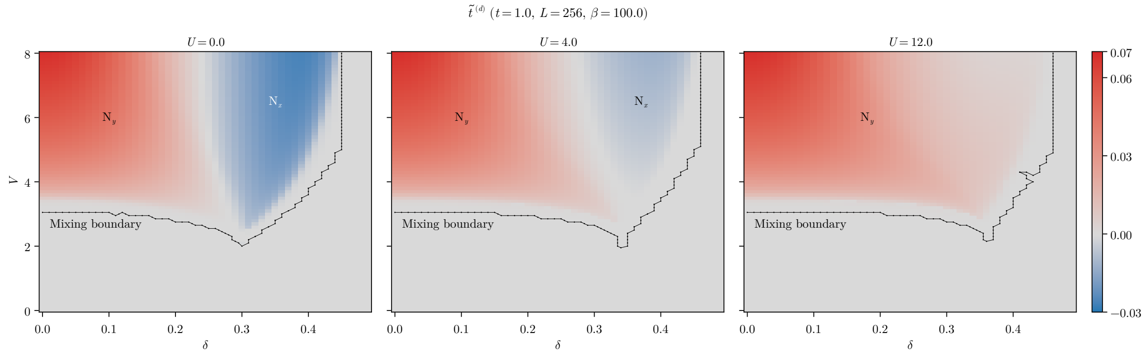


Figure 2.7 Heatmap of the anisotropic component of renormalised hopping, $\tilde{t}^{(d)}$, at $U/t = 0, 4, 12$ from left to right. The dataset is the same of Fig. 2.5, and the mixing boundary is the set of points separating the mixed region from the rest of the phases.

Nematic effects are directly related to symmetry mixing in the superconducting order parameter. Consider Fig. 2.7, where we plot the convergence values for $\tilde{t}^{(d)}$ in the (δ, V) plane for $U/t = 0, 4, 12$. The used dataset is the same of Fig. 2.5. The superimposed black lines are the mixing boundaries, the set of extremal points of the deeper blue regions of Fig. 2.5. Despite a bit of roughness related to the gridded nature of our data, it is evident that these boundaries encircle the regions where $\tilde{t}^{(d)} \neq 0$. Note that the set of points composing the mixing regions are those satisfying the threshold rule exposed in last section; choosing another threshold, the black curves change. Nevertheless, the key point is that there is an evident connection between symmetry mixing and breaking of planar rotational symmetry.

The mixing region coincides with the nematic region. As we said at the beginning of this section, nematicity can be established in two ways, based on the direction along which hopping is boosted. Hence it must hold

$$\mathrm{M} \simeq \mathrm{N}_x \cup \mathrm{N}_y$$

with the following convention on notation:

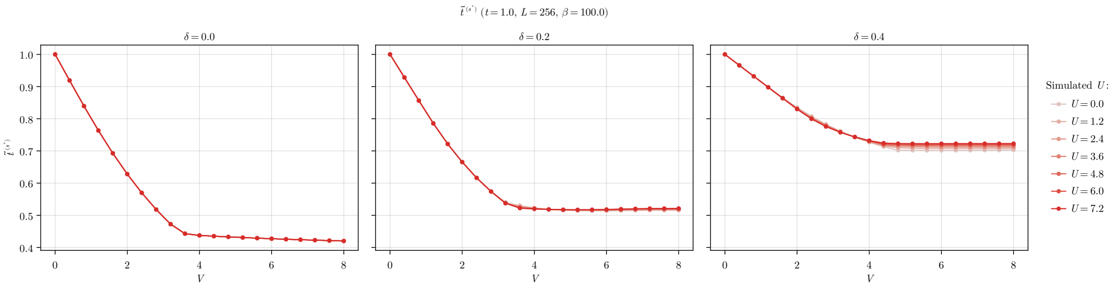
$$\begin{aligned} \mathrm{M} &= \text{Region of } (\delta, V) \text{ plane where mixing occurs} \\ \mathrm{N}_x &= \text{Region of } (\delta, V) \text{ plane where } \tilde{t}^{(d)} < 0 \\ \mathrm{N}_y &= \text{Region of } (\delta, V) \text{ plane where } \tilde{t}^{(d)} > 0 \end{aligned}$$

The notation N_ℓ has the following meaning: the index ℓ indicates the direction along which renormalised hopping is larger. We indicate in Fig. 2.7 these regions for $U/t = 0, 4, 12$.

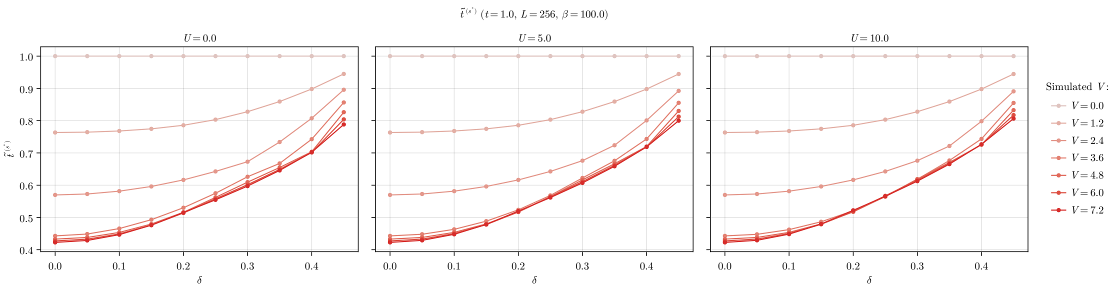
Note that the map were colour-corrected in order to enhance the visibility of the blue region. The intensity of the colouring there suggests that the amplitude is comparable with the red region, but is not. Indeed, we set up the colormap such that gray is aligned with $\tilde{t}^{(d)} = 0.0$, and intensity of colour grows asimmetrically in the negative and positive direction.

The three panels of Fig. 2.7 differ crucially in the fact that the data for $U/t = 12$ (right) are all positive, while those for $U/t = 0, 4$ (left,centre) are negative in a portion of space. In other words, as we increase U/t, N_y expands and N_x shrinks. Moving from $U/t = 0$ (left) to $U/t = 4$ (centre) we see that the amplitude of $\tilde{t}^{(d)}$ decreases in N_x and not in N_y. We already observed this in Fig. 2.6b. Moreover, for $U/t = 12$ (right) N_x is apparently absent and N_y takes up the entirety of the mixing region. Whether there exists some critical value of U/t above which N_x disappears in the entirety of the (δ, V) plane, we do not know.

Breaking of planar rotational symmetry and the insurgence of a d-wave harmonic for the renormalised hopping is certainly the most peculiar effect for the singlet SC phase. It adds to the isotropic bands shrinking, which is, the s^*-wave renormalisation of the hopping parameter. We discuss it in next section.



a | Dependency of $\tilde{t}^{(s^*)}$ on V at various Us, compared at three values of δ.



b | Dependency of $\tilde{t}^{(s^*)}$ on δ at various Vs, compared at three values of U.

Figure 2.8 | Dependency of the s^*-wave renormalised hopping $t^{(s^*)}$ varying the doping δ and the local repulsion U. In the top panel we compare three doping levels $\delta = 0.0,\ 0.2,\ 0.4$ varying $0 \leq U \leq 8$. In the bottom panel we compare three repulsion values $U = 0.0,\ 5.0,\ 10.0$ varying $0 \leq \delta \leq 0.45$.

2.5.3 Isotropic bands shrinking

The last HFP to be checked is $u^{(s^*)}$, responsible of bands shrinking. Its physical role is the isotropic renormalisation of the hopping parameter

$$\tilde{t}^{(s^*)} = t - \frac{V u^{(s^*)}}{2}$$

again, we skip plotting $u^{(s^*)}$ and plot directly $\tilde{t}^{(s^*)}$ in Fig. 2.8. In the top panel (Fig. 2.8a), we show the dependence of $t^{(s^*)}$ on V. The three curves, obtained at three doping levels, are approximately independent of U. The common feature is that $\tilde{t}^{(s^*)}$ decreases somewhat linearly with V up to some critical value, above which it flattens to an horizontal asymptote. Electronic doping changes primarily the height of this asymptote

As was for the AF and normal phases, electronic doping disrupts the isotropic bands shrinking effect. At half-filling (left figure) we see that bands shrinking is most effective, reducing the hopping parameter to less than half its $V = 0$ value. In the bottom panel (Fig. 2.8b) we show $t^{(s^*)}$ as a function of δ, varying V and U. As we already noted, the dependence on U is very weak and the three figures are a lot alike. As we already noted, $t^{(s^*)}$ increases with doping.

In Fig. 2.9 we compare $t^{(s^*)}$ for the normal phase (blue surface) and the SC phase (green surface). The two surfaces share a general feature: $t^{(s^*)}$ is approximately unitary at low V/t and almost unitary doping, and decreases significantly in the low-δ and high-V/t limit. In this region, for the normal phase we observe the Mott-localisation plateau discussed in Sec. 1.6.1, where $t^{(s^*)} \approx 0$. In the SC phase, looking at the same region, $t^{(s^*)}$ is minimised but no plateau is formed. More importantly, isotropic hopping is reduced to roughly half the value of t and not nullified. We do not observe a Mott-like area as for the normal phase.

Recalling the plots of Fig. 2.5, the low-δ and high-V/t limit (the corner where $t^{(s^*)}$ is minimised) falls deep within the mixed region. Recalling also Fig. 2.4, in this corner the d-wave and s^*-wave harmonics of the gap function are large and comparable. Finally, we know from Fig. 2.7 (central panel) that here nematic effects are small, but significantly non-zero. In particular, considering

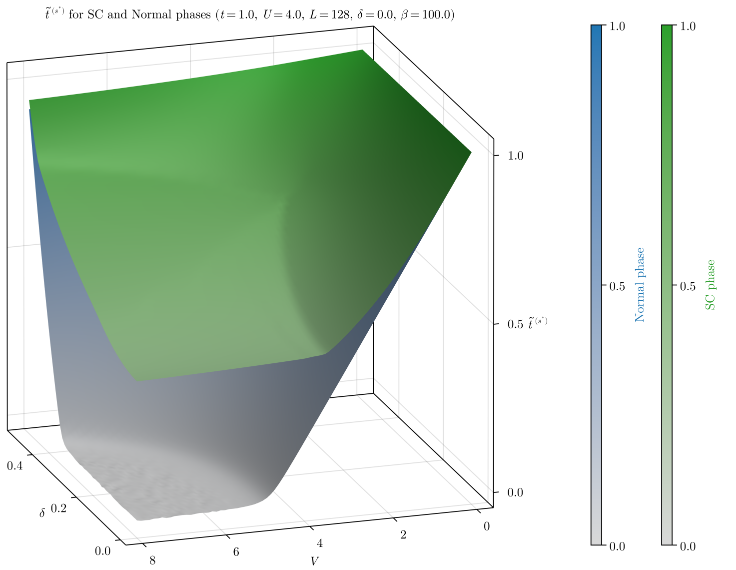


Figure 2.9 Comparison of the isotropic part of the renormalised hopping $\tilde{t}^{(s^*)}$ for the SC (green) and normal (blue) phases on the (δ, V) plane. The data from the normal phase are the same of Fig. 1.7.

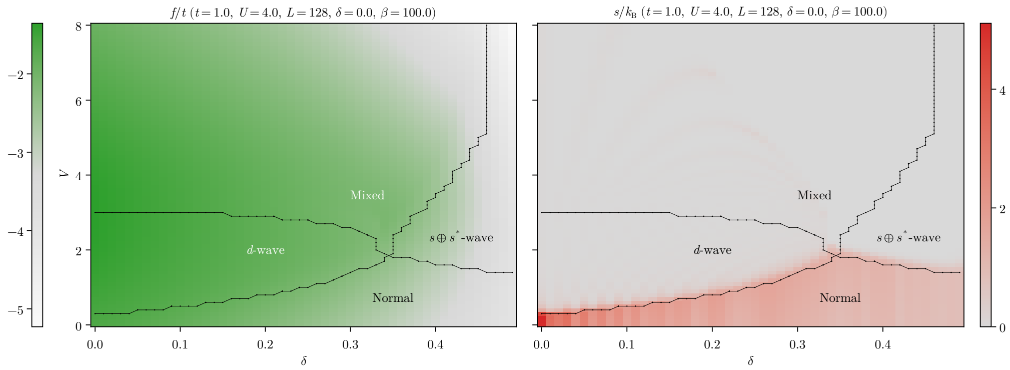


Figure 2.10 Heatmaps of the free energy and entropy densities at $U/t = 4$. The black dotted lines represent the internal boundaries of the singlet SC phase, extracted from Fig. 2.5.

that

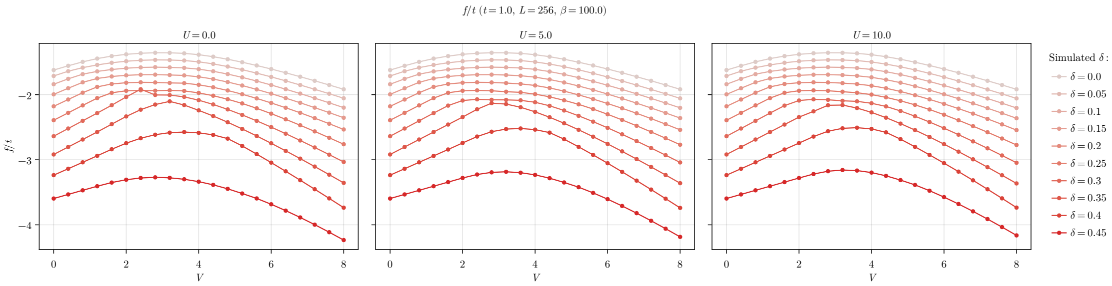
$$\lim_{\delta \to 0} \lim_{V/t \gg 1} \frac{\tilde{t}^{(s^*)}}{t} \approx 0.4 \qquad \lim_{\delta \to 0} \lim_{V/t \gg 1} \frac{\tilde{t}^{(d)}}{t} \approx 0.06$$

we see that the two components differ just by one order of magnitude. Nematic effects account for a rather large portion of hopping renormalisation.

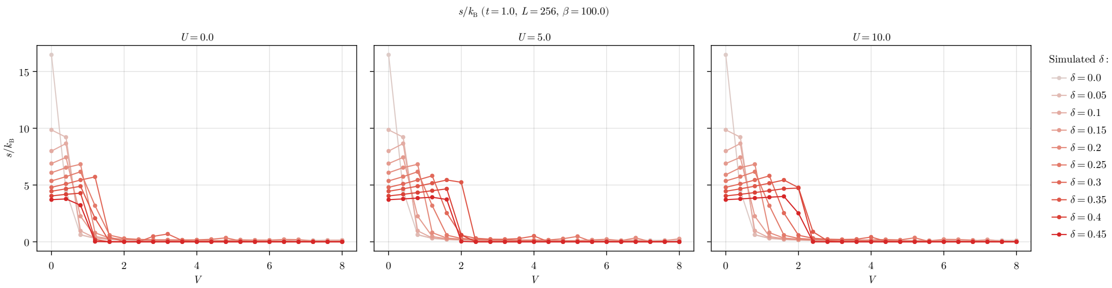
With this consideration, we end the presentation of HFPs and RMPs and move to the last section, where we discuss the results regarding free energy and entropy densities.

2.5.4 Solution free energy and entropy densities

For the SC phase, free energy density is given by Eq. (2.61), while entropy density is given by Eq. (2.60). In Fig. 2.10 we plot, sided, the heatmaps for f and s at $U/t = 4$ in the (δ, V) plane. Superimposed, we show the boundaries between the four regions identified in Fig. 2.5. Consider first the free energy, on the left: our data suggest that free energy is maximised in the d-wave phase, followed by the mixed phase. At high doping, the $s \oplus s^*$-wave phase is low-energetic. Overall, f varies smoothly across phase boundaries, and we do not identify the black lines with some isoenergetic line separating phases.



a | Dependency of f on V at various δs, compared at three values of U.



b | Dependency of s on V at various δs, compared at three values of U.

Figure 2.11 | Dependency of free energy (top) and entropy (bottom) densities on V varying the doping δ and the local repulsion U.

This is not the case for entropy density. Indeed, the entropic region on the right panel of Fig. 2.10 coincides with the normal phase. The three superconducting regions are low-entropic. This is explained by the following argument:

- In the normal phase, the Bogoliubov band reduces to $\tilde{E}_{\mathbf{k}} \to \tilde{\xi}_{\mathbf{k}}$. The chemical potential crosses the bare band and a Fermi surface exists. Hence, at finite temperature, the Fermi-Dirac occupation of single-particle states varies continuously from almost unitary (deep in the Fermi sea) to almost zero (far outside). All those states in the middle, around the Fermi level, are highly entropic: we have shown this in Fig. 1.10b. It is the intrinsic metallic nature of the normal phase that makes is entropic.

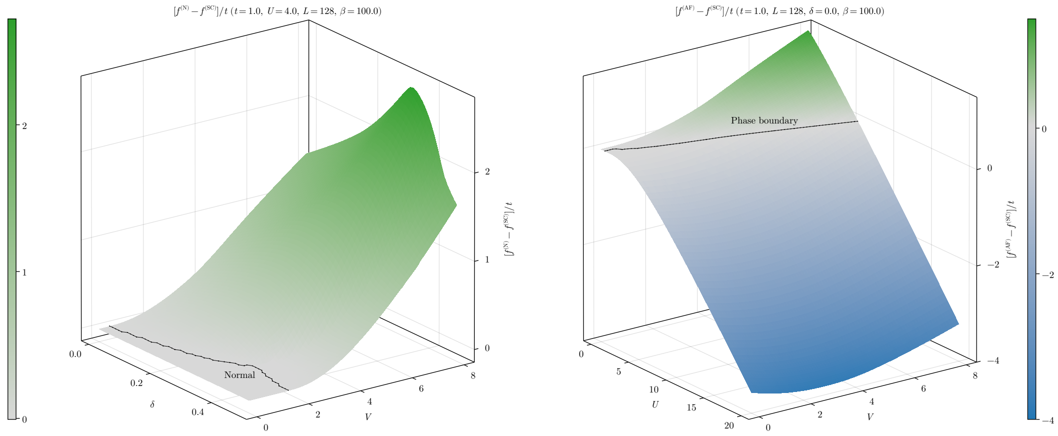
- In the SC phase, we open a gap at the chemical potential, not at zero energy (as instead was in the AF phase). Computing entropy as in Sec. 2.4.3, we apply the Fermi-Dirac distribution on the two bands $\pm\tilde{E}_{\mathbf{k}}$. As the gap opens, the lower one fills ($f \approx 1$), while the upper empties ($f \approx 0$). As we lower temperature, the above statement gets exact. Recalling Fig. 1.10b, we see that entropy is minimised at $f = 0, 1$. Hence, the SC phase is low-entropic.

This is the reason behind the sharp separation of the normal region from the SC regions in the right panel of Fig. 2.10.

In Fig. 2.11 we show f and s as functions of V, varying δ and comparing for three values of U. On both rows, moving from left to right we do not observe significant differences: f and s depend weakly on U. We already observed the presence of a maximum of the free energy at ratios V/t comparable with 1. In Fig. 2.11a we see this maximum around $V/t \lesssim 4$. Comparing with the data of Fig. 2.10, we can imagine that this maximum could be located at at the boundary between d-wave and mixed regions. Our data and methods are too rough to assert with certainty.

In the bottom panel, entropy density (Fig. 2.11b) shows the aforementioned sharp separation between normal and SC regions, and is maximised at half-filling and $V = 0$. That is the case of the conventional HM. We already argued that superconductivity is impossible at HF level, while commensurate antiferromagnetism is favoured at null doping with respect to the normal phase.

To conclude, we compare the free energy of the SC phase with that of the normal phase and the AF phase. We said that f depends rather weakly on U for the singlet SC phase, so we carry



a | Energy difference with normal phase. b | Energy difference with AF phase.

Figure 2.12 | Surface plots of the energies among the three phases. For the normal phase, we use that fact that $f^{(\mathrm{SC})}$ depends weakly on U, while $f^{(\mathrm{N})}$ does not depend on it at all, to limit to the case $U/t = 4$. For the AF phase, we limit to the half-filling condition and show data in the (U, V) plane.

out the comparison with normal phase at $U/t = 4$ in the (δ, V) plane. In Fig. 2.12a we plot the quantity

$$f^{(\mathrm{N})} - f^{(\mathrm{SC})}$$

and observe that it is greater or equal to zero over the entire explored region. The energy difference is null only at $V/t \lesssim 2$, where the superconducting gap is null. We plot, superimposed, the boundaries of the normal region extracted from Fig. 2.5. In the rest of the (δ, V) plane the SC phase is energetically favoured with respect to the normal phase.

In Fig. 2.12b we show the energy difference between AF and SC phases,

$$f^{(\mathrm{AF})} - f^{(\mathrm{SC})}$$

We colour-corrected the surface plot to align the gray with the $f = 0$ level. Colour separates positive and negative regions: the SC phase is dominant where $f^{(\mathrm{AF})} > f^{(\mathrm{SC})}$ (green region), in the rest of the plot the AF phase dominates (blue region). We indicate the contour level at $f = 0$ as a black dotted line, obtained as the set of data points whose energy difference is closest to zero. At half-filling, commensurate antiferromagnetism is almost everywhere energetically favoured. Singlet superconductivity prevails in the $U/t \ll 1$, $V/t \gg 1$ limit. The fact that the phase boundary has somewhat a linear shape suggests the fact that possibly a critical ratio exists,

$$\text{(Critical ratio)} \qquad \left(\frac{V}{U}\right)_{\mathrm{c}} = k$$

delimiting a phase transition from antiferromagnetic to superconducting ordering at half-filling.

2.6 A final note

We conclude this chapter with a note of method. We performed simulations on the singlet channel, but also gathered data for the triplet channel. We omitted to show them in order to avoid unnecessary complications. Indeed, we observed that triplet superconductivity (as we formulated it at HF level) is suppressed everywhere in the region of parametric space we are exploring. This means that

$$w^{(p_x)} = w^{(p_y)} = 0 \qquad \text{(Observed numerically)}$$

Although non present in this discussions, **the plots for these HFPs are openly accessible at our repository.**

We did not provide any theoretical constraint on the five HFPs for the singlet SC phase. We tried to obtain a proof similar to those presented for the normal and AF phases, but we did not find any. It would be interesting, if possible, to work out a method to express analytically the position of the various boundaries we identified.

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